How Shared Outputs Shape Source-Specific Evidence in In-Context Learning

Anonymous ACL submission

Abstract

When several sources demonstrate different rules using the same output vocabulary, how does a language model form and use evidence for the requested source? We distinguish the source positions accessed by a query from the origin of the learned relations influencing its answer. Earlier demonstrations can alter another source's evidence before the query is processed. Causal message interchanges identify an output-dependent pathway from historical answers to later label states, and a frozen reconstruction predicts namespace counterfactuals in new contexts. Removing the corresponding native operation reduces cross-source relational influence across model families. With output words fixed, the messages also transfer part of a learned input–label association. This formation account predicts a constructive distinction: selectively limiting cross-source answer formation can improve source-conditioned judgments while retaining other contextual access paths, whereas isolating all cross-source formation can worsen them. Query-stage isolation has different effects across models. On real sentiment text, the same formation mechanism holds with reliable private-rule controls, and selective formation improves shared-word judgments by 5.4–9.5 percentage points; criterion–convention counterfactuals further distinguish association transfer from full criterion-related responses. Information-equivalent public backgrounds can also produce private caches whose joint use differs, including when both sources share identical output words. The results connect evidence provenance, historical formation, and source-conditioned use, explaining why attention to the requested source alone is an incomplete account of evidence selection.

1 Introduction

A context can contain several legitimate rules. Two annotators may assign different labels to the same kind of comment, or judge different aspects of it, while sharing the words `yes` and `no`. A new query specifies whose demonstrations should govern its answer. The model must infer and use the relevant input–output relation, while retaining useful information from the surrounding context. This is a computational question both when its answer is correct and when the conventions interfere.

Mechanistic studies have established input encoding, merging at label positions, retrieval, and aggregation in ICL. Cross-demonstration contextualization is also important: examples can acquire task information from earlier examples before a query uses them. What remains to establish in the multi-source setting is how this process organizes **the provenance of learned relations**. Source A's token positions belong to A, but their contextualized representations may have been shaped by B's rule. The output vocabulary can influence that formation process, as well as the final answer.

This changes the object of explanation. A model attending to the requested source has selected positions, not necessarily isolated that source's supervision. Conversely, cross-source information is not uniformly harmful: removing it can discard information that helps the model use either rule. A computational account must explain which relations enter the available evidence and how the reading process uses that evidence.

We develop this account through three connected findings:

1. **Output coding affects evidence formation before retrieval.** Source-position access and relational provenance can be causally separated. A fixed historical answer-to-label intervention predicts output-namespace counterfactuals, and native operation removal establishes a functional contribution.
2. **The historical messages carry learned associations.** Counterfactuals preserving label words, positions, and input multisets distin-

guish relational content from a static output signal. These effects remain under full-source reading and on real sentiment text.

3. **Formation and reading impose different functional constraints.** A prospective comparison of selective formation, complete source separation, and query-only isolation produces different outcomes for two source-specific rules. Public source–output relations can also improve their joint use through formation while private histories remain isolated; direct query access gives different results. Equivalent public backgrounds yield different joint cache compatibility, including under identical output words. Criterion–convention counterfactuals show why association transfer does not by itself specify a full criterion program.

Our contribution builds on existing contextualization mechanisms. The increment is a tested account of output-dependent relational provenance and its consequences for source-conditioned evidence use. It does not rely on future models continuing to fail the task.

2 Measuring which source's relations influence a prediction

2.1 Sources, rules, and materials

The current sentiment and dual-aspect experiments contain eight demonstrations from each source and eight novel queries. In shared-word conditions both sources use `no/yes`, with private label polarity balanced across contexts; separate-word conditions keep A's words and give B `pass/flag`; earlier Measuring Hate Speech experiments use `safe/toxic`. A's raw examples and labels are held fixed; paired counterfactuals change B. A single-aspect sentiment task separates association from output naming. A dual-aspect task independently varies food and service sentiment, B's inferred criterion, and its private output convention. Original SST5 reviews supply a natural-text test with independently sampled demonstrations and official held-out queries.

All models are frozen: Qwen3-8B, Qwen3-14B, and Qwen3-32B post-trained hybrid checkpoints, Mistral-7B-v0.3 base, and Llama-3.1-8B base. We score the serialized task as a continuation using FP32 eager attention. Confirmation uses new materials after discovery, without selecting models, contexts, or seeds by correctness. Model revisions, actual token streams, and complete split rules are recorded in the appendix.

2.2 Positional access and functional provenance

Write the contextualized source caches and prediction as

$$H_A = F_A(D_A, D_B), \quad H_B = F_B(D_A, D_B),$$
$$z = R(q, H_A, H_B). \tag{141}$$

Here z is the difference between the two output-label logits (`yes` minus `no` in the current sentiment experiments). The notation describes intervenable quantities rather than separate neural modules. To measure B-relation dependence, we change B's input–label pairing while fixing A and the query. We also prevent **every query position** from accessing B positions. Any remaining B-rule effect must reach the prediction through the remaining contextualized graph.

Under this restriction, changing B's rule changes 17.6% [12.1, 23.8] of A's answers in 32 independent Qwen3-8B Measuring Hate Speech contexts (E90). Separating B's output vocabulary reduces the inherited effect even though A's raw demonstrations and labels are unchanged. Source identity also has native causal effects in earlier interventions. Thus the result concerns the provenance of contextualized evidence, rather than a model completely ignoring source names.

3 How output identities shape historical evidence

3.1 A frozen formation-path prediction

We compare shared-output and separate-output prefixes, keeping A's examples and output words identical. Counterfactual donor states replace historical B-answer-region K, V, or KV **only in messages consumed at A's label positions** during prefix processing. Subsequent states evolve normally. Interchanging the resulting A-label caches into one otherwise pristine recipient tests the specified formation interface.

The prediction has coefficient one: the message-intervention effect predicts the actual namespace counterfactual at that interface. No head, layer, strength, or regression coefficient is selected. We report raw effects and rule double contrasts, which subtract the namespace effect between two

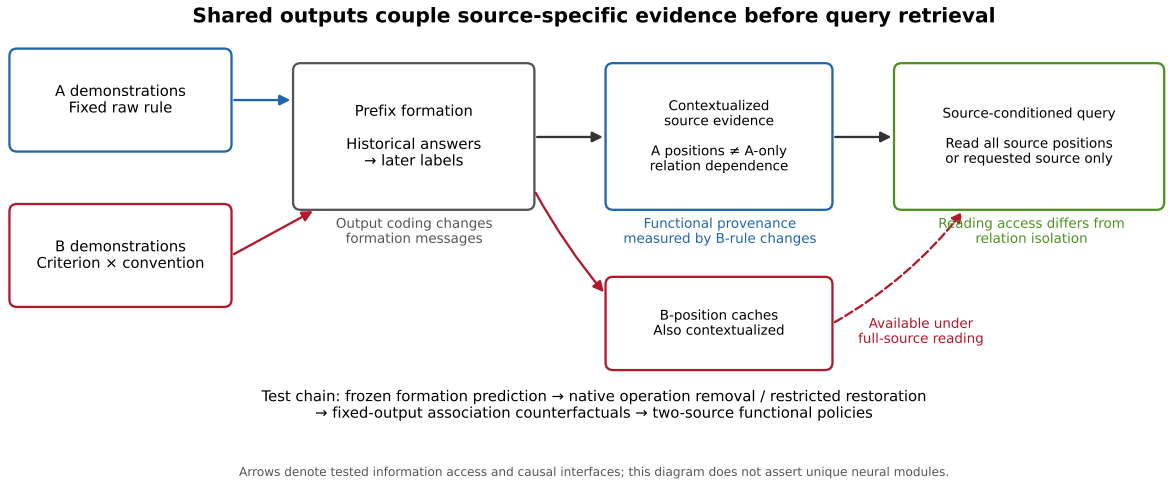


Figure 1: Historical formation creates contextualized evidence for both sources. Changing B can affect evidence at A positions, while full-source reading can also access B-position caches. Arrows denote tested causal interfaces and information access.

B-association worlds. This separates relation-dependent naming effects from a common output shift.

In new contexts, K reconstruction gives rule-double-contrast correlations .969 and .966 for Qwen3-8B and Mistral on the controlled Measuring Hate Speech task; KV gives .985 and .977. The same frozen mechanism remains predictive on controlled sentiment and real SST5 text. These are actual-donor causal reconstructions: their target is a specified interface effect, and residuals are measured against the coefficient-one identity prediction.

3.2 Testing the native operation

Reconstruction alone does not establish native contribution. We therefore block historical B-answer access at A-label receivers during ordinary prefix processing, using the same fixed mask at every layer. There is no donor activation. A complementary test blocks all B-to-A prefix access, then reopens only the historical answer-to-label path. Within-A answer removal and single-source private functions control for disturbance to general historical learning.

For each context, the signed B-rule response is the query mean of A-correct-label-oriented margins, $D = \mathrm{mean}[g_A(q)(z_{\mathrm{cross}}(q) - z_{\mathrm{origin}}(q))]$. Removal reports $D_{\mathrm{native}} - D_{\mathrm{blocked}}$; restricted restoration reports $D_{\mathrm{restore}} - D_{\text{all-B-off}}$. The targeted native removal reduces B-rule influence in every checkpoint. Reopening the path in the restricted graph increases influence under A-only reading, but has negative, positive, or uncertain effects under full-source reading. Thus its function depends on the subsequent evidence environment. Masks renormalize attention and later states remain live; the experiment identifies the operation's conditional contribution rather than a fixed additive message fraction. In Mistral, removing its own answer history also impairs its private function, and Qwen's cross-versus-within specificity interval includes zero. Complete controls remain in the appendix.

4 The messages transmit relationships, not only label identity

Namespace changes might alter output-related signals without changing learned relational content. We therefore hold B's label stream, source names, positions, and input multiset fixed, and reassign comments across opposite sentiment classes. A matched within-class reassignment moves the same number of inputs without reversing the association. A's raw evidence and novel queries stay fixed; paired token geometry is matched before model scoring.

For each world, we measure the correct-label-oriented query effect, then compare its magnitude between cross-class and within-class reassignment. Historical answer KV has an excess effect of .618 [.446, .813] nats in Qwen and .404 [.313, .499] in Mistral under A-only reading. V carries substantial content, while the namespace reconstruction is more accurately reproduced through K/KV. This dissociates the intervention's relation content from

Model, n=32	Full-source operation removal	A-only restricted restoration	Full-source restricted restoration
Qwen3-8B	1.649 [1.108, 2.250]	1.455 [0.999, 1.945]	-0.806 [-1.365, -0.266]
Mistral-7B-v0.3	1.054 [0.886, 1.227]	0.647 [0.463, 0.835]	0.140 [0.053, 0.231]
Llama-3.1-8B	0.808 [0.517, 1.109]	0.958 [0.635, 1.276]	-0.233 [-0.336, -0.122]
Qwen3-14B	1.329 [0.967, 1.717]	1.120 [0.673, 1.646]	0.039 [-0.217, 0.287]

Table 1: Native operation removal and restricted restoration, n32 confirmation. Signed B-rule contrasts in nats with context-bootstrap 95% CIs.

its output-identity effect without assuming independent internal components.

The relation effect also holds when queries can read both sources. Moreover, crossing A-position caches formed under shared/separate namespaces with B-position caches from the same backgrounds changes native B-rule dependence. The signed interaction is positive in the three smaller checkpoints and negative in Qwen3-14B ($-.686$ [-1.133, $-.237$] nats). The model therefore consumes jointly formed evidence; its source caches are not independent stores whose effects can be assumed additive.

The answer-region pathway reproduces part of the relation effect. Full label interchanges have additional effects, and the signed relation contrasts vary by model. These distinctions motivate measuring complete functional behavior in the next experiment, rather than equating a message effect with a fully transferred rule.

5 A constructive test: preserve relations while retaining context

5.1 Competing formation and reading policies

The formation account predicts different consequences for three operations: selectively limiting other sources' answer messages, removing all cross-source formation, and restricting only query access. The first tests whether limiting answer sharing protects private mappings while retaining other contextual paths; the second can also remove useful shared information; the third acts on evidence already formed.

We fix three prefix policies—native, symmetric other-source answer removal at label receivers, and all cross-source removal—and two whole-query scopes. Both source requests use the same prefix. On dual-aspect comments, B's food/service criterion and same/opposite private convention vary independently. The primary averages two-candidate accuracy over both sources, worlds, and queries within each context. Single-source functions and a one-sentence instruction are evaluated separately.

5.2 Independent functional confirmation

Selective formation improves mean accuracy by 2.93 [1.42, 4.35], 3.42 [1.17, 5.76], 7.37 [5.37, 9.43], and 2.78 [0.49, 4.79] points in Qwen3-8B, Mistral, Llama, and Qwen3-14B. It outperforms complete cross-source formation removal in every checkpoint. Gains concentrate on conflicting correct labels; agreeing-label cases lose 2.34 [0.78, 4.00] points in Qwen3-8B and 3.61 [1.56, 5.86] in Qwen3-14B. Individual-source gains are not uniformly confirmed. Query-only isolation gives larger gains in Llama and Qwen3-14B, so selective formation is not a universally optimal policy.

Formation isolation and query isolation control different routes: complete formation isolation can increase convention influence when the query reads both sources, whereas combining it with source-only reading gives the graph-required zero B-world effect on A. These are constructive policy tests with known source ownership. Masks remove unequal edge sets, renormalize attention, and leave subsequent states live; the comparison measures functional consequences rather than an equal-mass isolated component. Appendix C retains every policy, source, agreement tradeoff, and private-function control.

5.3 Association influence and criterion-related function

Source-conditioned use also depends on what a learned association does to a new input. To test this, we independently reverse B's private convention under food and service criteria, holding A fixed. Responses are projected onto food, service, bias, and their interaction. The resulting two-by-two criterion–feature matrix distinguishes changing the amount of influence from changing its functional direction.

In Qwen3-14B, native food and service selectivity are .282 [.199, .371] and .404 [.294, .528]

nats. A prospective within-context normalized determinant is .410 [.330, .493], rejecting a restricted account in which both criteria only rescale one common functional direction. With the query prohibited from reading B, it is still .190 [.073, .310]. Local label and answer-message interfaces yield different responses. Historical answer KV only partially reconstructs the **actual label-interface** criterion counterfactual (r=.668, RMSE=.646, zero RMSE=.746).

Thus association transfer, full-history criterion dependence, and correct source-conditioned use are distinct measurable properties. The complete response matrices and negative effects are retained in the appendix; a positive aspect response alone is not sufficient evidence for a complete abstract criterion program.

6 Natural-text formation with usable private rules

6.1 Mechanistic replication with reliable private conventions

We repeat the frozen 43-condition engine on original extreme-polarity SST5 reviews. Demonstration pools are disjoint text-hash partitions of training; discovery and confirmation queries use official validation and test splits. Gold sentiment and token geometry determine construction, without screening model outputs. Positive and inverted private conventions have single-source accuracies of 88.3–99.2% across the four checkpoints (Appendix D).

The same coefficient-one KV formation reconstruction gives rule correlations .931/.948/.930/.966 in Qwen3-8B, Mistral, Llama, and Qwen3-14B, respectively. Fixed-output association excess magnitudes are positive in every family. Llama's signed relation contrast is negative ($-.109$ [$-.194$, $-.024$] nats): transmitted content is not uniformly a correction of the requested rule. SST5 provides a functionally usable natural-text replication; these constructed conventions do not recover individual human preferences.

6.2 A prospective test of real-text source-function use

We next test the functional policy prediction on new SST5 contexts. A and B infer the same sentiment criterion but can adopt opposite private conventions. Origin and cross-convention worlds receive equal primary weight; a within-class movement remains a separate control. The same three formation policies and two query scopes are evaluated for both sources from each common prefix. A keeps `no/yes`; B uses either those shared words or `pass/flag`. Single functions and one-sentence source instructions are retained in all 85 conditions.

Selective formation improves the shared-word primary in all four checkpoints, with independently positive A and B gains. Candidate and full-vocabulary primary scores coincide here. Benefits concentrate on opposite conventions; same-convention effects are small and uncertain. The predefined B-convention response also decreases under selective formation while A's output words remain fixed (Appendix F), connecting the accuracy change to relational influence. Query-only isolation remains more effective in Llama and less effective in the two Qwen checkpoints; Mistral's point ordering is uncertain.

Separate words expose a further distinction. Complete cross-source formation isolation retains about 94–97% own-word candidate accuracy, yet full-vocabulary accuracy falls to about 51–54%: the model frequently outputs the other source's word. Thus learning within a label space and using that convention jointly require separate measurements. Native separate-word candidate accuracy is already high, so its smaller selective gains also reflect headroom and word priors. Mistral loses 1.27 [0.20, 2.44] full-output points under selective formation despite unchanged candidate accuracy. Appendix F preserves every namespace, private direction, source, actual output, and instruction control.

6.3 Sharing identities while keeping supervision private

What can be shared without exchanging private supervision? With cross-source demonstration formation blocked, we cross access to an unlabeled source-name list during formation and whole-query reading, using unrelated names as control. Qwen3-8B has a formation-specific full-output gain of 4.59 [2.73, 6.64] points, with uncertain candidate gain; Mistral's effect is negative and Llama's uncertain (Appendix G). Names alone therefore provide a partial, model-dependent account of the missing shared information.

6.4 Sharing an output schema before private rule learning

We next share a source-to-word-set table before independent formation, for example Alex: no, yes;

Model, confirmation n=32	Shared native accuracy, %	Selective formation, %	Selective-minus-native, pp [95% CI]	Query-only-minus-native, pp [95% CI]
Qwen3-8B	84.86	92.19	7.32 [5.37, 9.28]	-0.39 [-3.12, 2.44]
Mistral-7B-v0.3	88.77	94.14	5.37 [2.93, 7.71]	5.96 [3.22, 8.59]
Llama-3.1-8B	81.25	90.72	9.47 [7.52, 11.52]	16.70 [14.36, 19.04]
Qwen3-14B	89.94	97.36	7.42 [5.57, 9.38]	3.81 [1.27, 6.25]

Table 2: Real-text private rules with shared words: policy consequences, confirmation n32. Percent accuracy and paired changes with context-bootstrap 95% CIs.

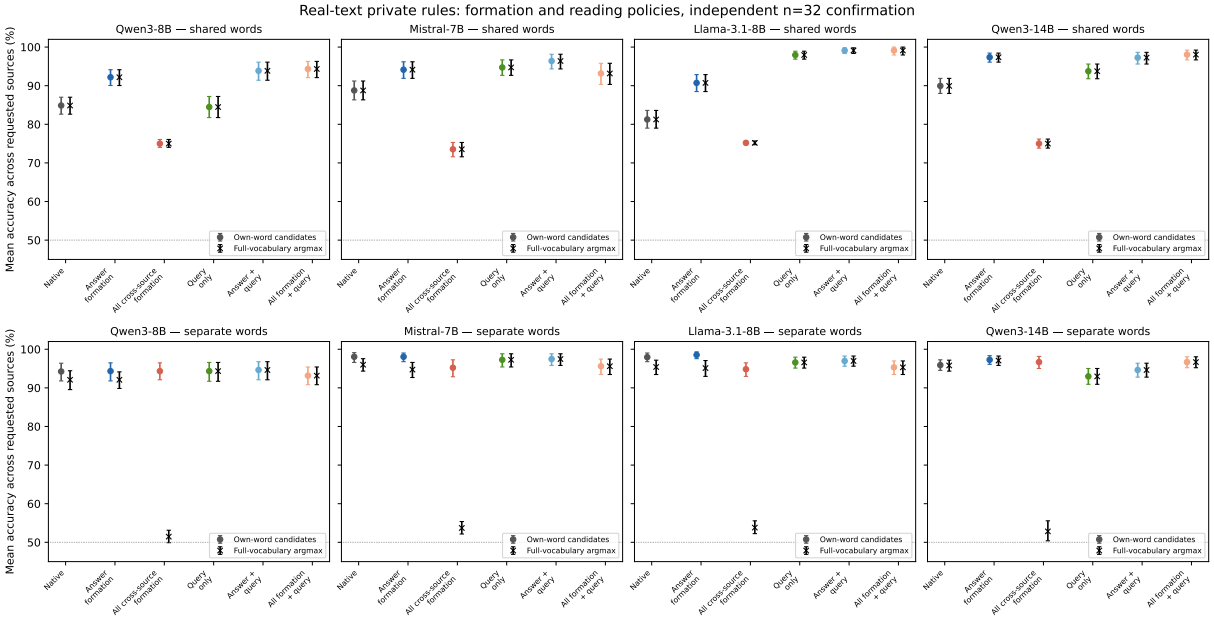


Figure 2: All formation/query policies on independently confirmed real-text private functions. Colored circles score within the requested source's two words; black crosses score the actual full-vocabulary output. The separate-word condition reveals failures hidden by candidate restriction.

Sam: pass, flag. It specifies output spaces, without review–label pairs or private polarity. An unrelated-name table has the same words and structure; a swapped table reverses the real sources' word assignments. Formation and query access are crossed while all private cross-source formation remains blocked.

True-minus-unrelated formation-only gains in full-vocabulary accuracy are 26.66 [23.05, 30.47], 7.71 [5.08, 10.45], and 8.30 [6.05, 10.84] points in Qwen3-8B, Mistral, and Llama. Direct reading-only contrasts include zero in all three. Public relational information therefore affects jointly usable private evidence through formation, even when no query position can read it.

Literal table routing is insufficient: correct tables outperform swapped tables, but swapped tables themselves improve full accuracy and reduce other-source outputs in every checkpoint. Qwen's candidate accuracy also falls, and some source-specific gains remain uncertain (Appendix H). The result identifies useful shared formation structure, with costs; the next test asks whether information-equivalent structures form equally compatible evidence.

6.5 Common formation backgrounds constrain joint rule use

We keep every public relation and private demonstration fixed, reversing only the two rows of the correct schema. Private cross-source formation remains blocked, and no query position can read the table. Each ordering supplies the same facts but forms its own pair of source caches. We combine A and B caches from one ordering (00 or 11), or from different orderings (01 or 10), interchanging all private K/V states at unchanged global positions.

With separate word sets, coherent composition improves full-output use much more than within-source candidate judgments (E126; Appendix I). A post-hoc diagnosis finds that most wrong-source words also match that other source's private con-

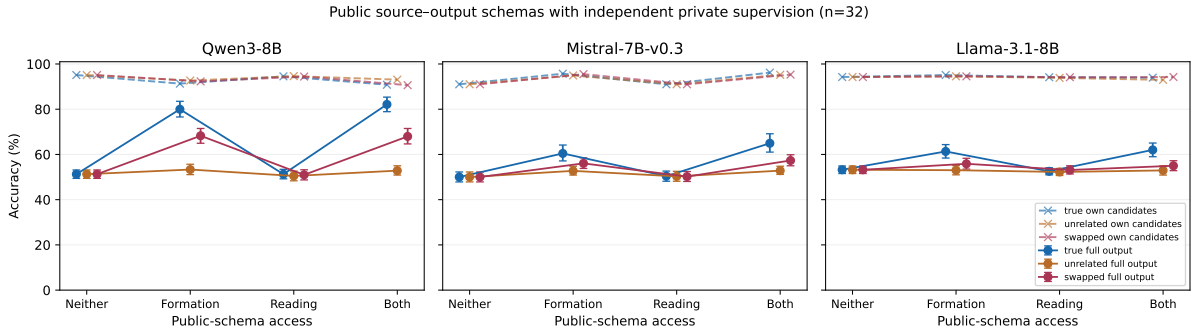


Figure 3: Public source–word-set access with private historical supervision isolated. Solid curves score the actual full-vocabulary output; dashed curves score within the requested source's candidates. Correct and swapped relations both help through formation, but their magnitudes differ.

vention. This motivates a prospective test with identical output words: a model must select the appropriate learned mapping even though the words themselves cannot identify a source.

E127 fixes four checkpoints and both shared/separate namespaces before scoring. Its primary uses shared words and opposite private conventions, where the two sources require different answers. The coherent-minus-mixed contrast averages 00 and 11 against 01 and 10. It cancels additive differences in individual-cache quality, while retaining the joint reader's nonlinear behavior.

Coherent composition improves shared-word, opposite-convention accuracy by 8.89 [6.64, 11.13], 5.86 [4.10, 7.71], 0.78 [0.10, 1.56], and 9.38 [7.32, 11.52] points in Qwen3-8B, Mistral, Llama, and Qwen3-14B. The source-conditioned margin also increases in each checkpoint (Appendix J).

The same dependence therefore extends beyond selecting a distinguishable word set. The paired source-conditioned margin is one half of the gold-A-oriented difference between A and B requests. A single request-independent output shift cancels in this difference and cannot explain its positive change. This is an equivalent projection of the two-source margins, not an additional independent mechanism measure.

The caches retain strong requested-source-only functions, yet coherent joint-reading mean accuracy under conflicting conventions is only 53–65%. The small Llama effect and nonuniform private-direction gains remain reported. With food/service criteria instead of opposing codes, coherent formation yields a smaller gain: Qwen3-14B improves by 2.93 [0.59, 5.47] points, while the other accuracy intervals touch or cross zero; joint-reading mean accuracy is about 50–53% (Appendix K).

These comparisons show that local private-function accuracy does not determine joint usability. The shared-word result exceeds a word-set-only explanation, and the source-request margin excludes a single request-independent output shift. It remains compatible with contextual binding and competitive retrieval. Whole-cache interchanges do not identify a new binding ID or a complete abstract criterion program; all cache pairs, common shifts, and private controls remain in Appendices I–K.

6.6 The same formation and operation chain in a larger model

We repeat the fixed formation, native-operation, and real-text function tests in Qwen3-32B, with a fresh Qwen3-8B confirmation anchor. The two sizes receive byte-identical contexts within each stage. All 43/64/85 conditions, K/KV/V channels, reading backgrounds, private directions, and instructions are retained. Discovery uses n16 and confirmation n32 without selecting by discovery effects. Full FP32 weights and eager attention are kept, with no fitted head, layer, intervention strength, or reconstruction coefficient.

The 32B formation batch has 100% single-source accuracy under both private directions, while the historical relation-formation effect remains. Coefficient-one label-interface KV reconstruction has raw r=.883 and rule-double-contrast r=.922, with raw RMSE .864 versus zero-prediction RMSE 1.857. Native restricted restoration is positive under A-only reading (+.775 [.321, 1.284] nats) and negative under full-source reading (−.544 [−.833, −.276]), as in the fresh 8B anchor. The same formation operation therefore contributes

Qwen3, matched confirmation n32	A-formation signed, nats	Native operation removal, nats	Selective-minus-native full, pp	Query-only-minus-native full, pp
8B	2.351 [1.857, 2.837]	1.762 [1.264, 2.294]	7.62 [5.18, 9.96]	−2.54 [−6.54, 1.46]
32B	2.676 [2.266, 3.096]	1.972 [1.467, 2.500]	7.71 [5.76, 9.57]	5.86 [4.20, 7.52]

Table 3: The same formation, native-operation, and real-text function chain across model sizes. Within-family matched confirmation contexts, n32; full-output accuracy and signed native effects.

in the larger model, with its function depending on the subsequent reading environment.

On real-text shared-word functions, 32B native full accuracy is 87.99%, rising to 95.70% with selective formation; its advantage over complete cross-source formation removal is 20.41 [18.16, 22.36] points. The paired selective-gain difference is .10 [−3.22, 3.32] points, while the query-only gain differs by 8.40 [3.81, 13.09]. Formation and reading contributions can thus vary separately across these checkpoints. Their training recipes also differ, so the paired comparison characterizes computation across sizes rather than estimating a parameter-count scaling law. Both requested-source gains, agreement tradeoffs, separate-word output failures, and all reconstruction targets are retained in Appendix M.

7 Related work and positioning

Cho's circuit (Cho et al., 2025), information removal (Cho et al., 2026), Contextualize-then-Aggregate (Bakalova et al., 2025), and How Few-Shot Examples Add Up (Wang et al., 2026) establish input merging, information selection, contextualization, and QK/V contributions. Sun et al. (Sun et al., 2025) trace input–label mappings, including 70B/72B checks. These explain important components of our setting. We add a measurement of whose learned relations enter evidence, an output-dependent formation prediction, and its native and functional consequences.

Local task vectors (Zheng et al., 2026) and just-in-time representations (Li et al., 2025) establish heterogeneous task states. Feng and Steinhardt (Feng and Steinhardt, 2024), Dai et al. (Dai et al., 2024), and Mixing Mechanisms (Gur-Arieh et al., 2026) explain binding, ordering, and competing retrieval mechanisms; Test, then Route (Thavarasa and Sukumar, 2026) separates predicate testing from token-dependent routing. Our relational-content and compatibility tests do not establish a new binding component. Parallel Context Windows (Ratner et al., 2023) and CacheBlend (Yao et al., 2025) already address independently computed prefixes and KV context dependence. Our counterfactual fixes public facts and private positions, blocks private supervision exchange, and tests joint function under equivalent formation backgrounds.

Task recognition/learning (Yang et al., 2026) separates label-space alignment from within-space mappings; Li et al. (Li et al., 2024) test task composition and label remapping. Demonstration Conflict (Jiao et al., 2026) studies corrupted demonstrations, and Jurisdiction (lin Li et al., 2026) studies contextual authority over prior knowledge. Here both inferred source rules are legitimate; this motivates our question without establishing novelty by setup alone. Yu and Ananiadou (Yu and Ananiadou, 2024), Crosbie and Shutova (Crosbie and Shutova, 2025), and Racing Thoughts (Lepori et al., 2025) motivate operation-specific causal tests. Lepori et al. (Lepori et al., 2026) and CoSToM (Li et al., 2026) separate internal information from deployment; our frozen-model functional tests do not claim this general distinction first.

8 Reproducibility

Intervals use 10,000 context-bootstrap samples; queries within a context are not independent units. All predefined conditions, private directions, negative results, and reconstruction channels remain reported. We check actual token/mask consumption, full-token/cached scores, self interventions, graph-zero controls, and pristine caches. Independent sessions recompute raw quantities and graph/gold layouts before reading author summaries; these are independent analyses, not GPU reruns. Cards, script hashes, revisions, small summaries, figures, and audits are versioned. Raw JSONL and weights remain at the documented asset paths; complete interfaces and serialization are in the appendix.

9 Discussion

Source-conditioned ICL reads evidence whose relations were formed across a contextual history. Selecting the requested source's positions therefore

does not establish the provenance of the supervision influencing its answer. Shared output identities shape this historical formation: the pathway transmits learned relations, predicts namespace counterfactuals, and contributes to ordinary predictions. Its functional contribution changes with subsequent reading, including in the larger checkpoint.

This account produces constructive predictions. Selectively limiting answer-level sharing can preserve private judgments, while removing all historical sharing can impair joint use. Public source–output relations help through formation even when private supervision cannot cross sources and the query cannot read the public table. Information-equivalent formation backgrounds also change compatibility under shared output words. Thus understanding relevant evidence requires explaining both the formation of private associations and their coordination during reading. The question remains meaningful when a model answers correctly.

10 Limitations

The experiments cover three families, including 14B and 32B same-family checkpoints. They do not isolate parameter count from training recipe. Dual-aspect templates fix food before service; their responses do not establish a position-independent criterion program. Pre-input source-cue transfer does not reliably deploy both criteria despite usable private functions (Appendix L); small responses and other native task-state routes remain possible.

Cache interchanges and renormalized masks test specified interfaces with unmodeled paths; policy masks remove unequal edge sets. Reconstructions use actual donor activations. Intervals are not corrected for all comparisons. Private conventions are constructed, and SST5 sentiment may occur in pretraining. Earlier practical evaluations found limited accuracy consequences of source mixing. These constraints preserve the demonstrated formation and use results while limiting applications and full-algorithm claims. The technical appendix retains all controls and the claim ledger records registered scope.

References

Aleksandra Bakalova, Yana Veitsman, Xinting Huang, and Michael Hahn. 2025. Contextualize-then-aggregate: Circuits for in-context learning in gemma-2 2b. *Preprint*, arXiv:2504.00132.

Hakaze Cho, Mariko Kato, Yoshihiro Sakai, and Naoya Inoue. 2025. Revisiting in-context learning inference circuit in large language models. *Preprint*, arXiv:2410.04468.

Hakaze Cho, Haolin Yang, Gouki Minegishi, and Naoya Inoue. 2026. Mechanism of task-oriented information removal in in-context learning. *Preprint*, arXiv:2509.21012.

Joy Crosbie and Ekaterina Shutova. 2025. Induction heads as an essential mechanism for pattern matching in in-context learning. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 5049–5111, Albuquerque, New Mexico. Association for Computational Linguistics.

Qin Dai, Benjamin Heinzerling, and Kentaro Inui. 2024. Representational analysis of binding in language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17468–17493, Miami, Florida, USA. Association for Computational Linguistics.

Jiahai Feng and Jacob Steinhardt. 2024. How do language models bind entities in context? *Preprint*, arXiv:2310.17191.

Yoav Gur-Arieh, Mor Geva, and Atticus Geiger. 2026. Mixing mechanisms: How language models retrieve bound entities in-context. *Preprint*, arXiv:2510.06182.

Difan Jiao, Di Wang, and Lijie Hu. 2026. Understanding the dynamics of demonstration conflict in in-context learning. *Preprint*, arXiv:2603.04464.

Michael A. Lepori, Tal Linzen, Ann Yuan, and Katja Filippova. 2026. Language models struggle to use representations learned in-context. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14841–14857, San Diego, California, United States. Association for Computational Linguistics.

Michael A. Lepori, Michael Curtis Mozer, and Asma Ghandeharioun. 2025. Racing thoughts: Explaining contextualization errors in large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3020–3036, Albuquerque, New Mexico. Association for Computational Linguistics.

Jiaoda Li, Yifan Hou, Mrinmaya Sachan, and Ryan Cotterell. 2024. What do language models learn in context? the structured task hypothesis. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12365–12379, Bangkok, Thailand. Association for Computational Linguistics.

Mengfan Li, Xuanhua Shi, and Yang Deng. 2026. CoS-ToM: Causal-oriented steering for intrinsic theory-of-mind alignment in large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9302–9317, San Diego, California, United States. Association for Computational Linguistics.

Yuxuan Li, Declan Campbell, Stephanie C. Y. Chan, and Andrew Kyle Lampinen. 2025. Just-in-time and distributed task representations in language models. *Preprint*, arXiv:2509.04466.

Pei lin Li, Qingle Liu, Junyang Feng, Siyu Li, Sunqi Fan, Xin-Sheng Chen, and Shuojin Yang. 2026. When context misleads: In-context learning with jurisdiction in large language models. *Preprint*, arXiv:2609.27603.

Nir Ratner, Yoav Levine, Yonatan Belinkov, Ori Ram, Inbal Magar, Omri Abend, Ehud Karpas, Amnon Shashua, Kevin Leyton-Brown, and Yoav Shoham. 2023. Parallel context windows for large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6383–6402, Toronto, Canada. Association for Computational Linguistics.

Chenghao Sun, Zhen Huang, Yonggang Zhang, Le Lu, Houqiang Li, Xinmei Tian, Xu Shen, and Jieping Ye. 2025. Interpret and improve in-context learning via the lens of input-label mappings. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3873–3895, Vienna, Austria. Association for Computational Linguistics.

Luxshan Thavarasa and Sivasuthan Sukumar. 2026. Test, then route: How language models execute in-context conditional rules across models and languages. *Preprint*, arXiv:2608.04183.

Entang Wang, Yiwei Wang, Aleksandra Bakalova, and Michael Hahn. 2026. How few-shot examples add up: A causal decomposition of function vectors in in-context learning. *Preprint*, arXiv:2605.16591.

Haolin Yang, Hakaze Cho, and Naoya Inoue. 2026. Localizing task recognition and task learning in in-context learning via attention head analysis. *Preprint*, arXiv:2509.24164.

Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang. 2025. Cacheblend: Fast large language model serving for rag with cached knowledge fusion. *Preprint*, arXiv:2405.16444.

Zeping Yu and Sophia Ananiadou. 2024. How do large language models learn in-context? query and key matrices of in-context heads are two towers for metric

learning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 3281–3292, Miami, Florida, USA. Association for Computational Linguistics.

Bowen Zheng, Ming Ma, Zhongqiao Lin, and Tianming Yang. 2026. Label words as local task vectors in in-context learning. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6319–6332, San Diego, California, United States. Association for Computational Linguistics.

A Complete criterion–convention functional response

A.1 Separating criterion from private output convention

A learned association can change a classifier's confidence without specifying a complete criterion for a new input. We test this distinction with comments containing independently balanced food and service sentiment. Source A's demonstrations establish a food criterion. Source B can evaluate either food or service, under an output convention agreeing with or opposite to A's. The criteria are inferred from demonstrations rather than stated in the multi-source prompt (E117/E118).

The four B worlds are constructed by bijectively reassigning inputs to fixed demonstration positions. A's raw demonstrations, B's output stream, source names, positions, and input multiset remain unchanged. Eight novel queries cover all four sentiment combinations. Every interface is evaluated on the same queries: complete native histories, histories whose queries cannot read B positions, A-label cache interchange into one recipient, and historical B-answer messages consumed during A-label formation. Separate-output label donors test the effect of output coding during formation while restoring the shared-output recipient for querying.

Let $f_q, s_q \in \{-1, +1\}$ be food and service polarity, oriented to A's private convention. For criterion t, define $\delta_t(q) = \frac{1}{2}[z_{t,+}(q) - z_{t,-}(q)]$, the response to changing B's convention. The context-level matrix

$$M_c = \begin{pmatrix} \mathbb{E}_q[\delta_{food}(q) f_q] & \mathbb{E}_q[\delta_{food}(q) s_q] \\ \mathbb{E}_q[\delta_{service}(q) f_q] & \mathbb{E}_q[\delta_{service}(q) s_q] \end{pmatrix}$$

separates the two criteria's effects on the two aspects. Bias and food–service interaction are also retained. These are B-relation effects while A is requested, not B's own accuracy or a gain in correctly isolating A.

A.2 Association magnitude and criterion selectivity give different answers

Across three model families, E117 confirmation finds that changing B's pairing alters both food and service sensitivity through A-label caches. A positive service response alone therefore cannot establish transport of B's service criterion. E118 independently crosses criterion with B's convention and preserves the entire response matrix. Full native histories show more criterion selectivity than local label interchanges in several settings, with substantial family variation. The mean criterion–feature interaction is positive in all three families, but Qwen3-8B's interaction mainly reflects service selectivity; unequal gains along one common direction can also produce a positive mean interaction.

Separating B's output namespace during formation can lower both aspect responses without appreciably changing their mean selectivity contrast. In Llama, food-criterion responses decrease by .267/.243 nats on food/service, and service-criterion responses by .110/.079; all four paired intervals are negative. Namespace effects on evidence formation thus include a change in relational influence magnitude, not only which aspect dominates.

A.3 A prospective functional-direction test in Qwen3-14B

Before any E119 scores, we freeze an additional observer: the determinant of each context's response matrix and its normalized determinant, $\det(M_c)/(\|M_{c,food}\|\|M_{c,service}\|)$. If the two criteria merely rescale a common functional direction, each determinant is zero. These quantities are computed within context before averaging. A row norm of at most .001 nats sets the normalized statistic to zero while retaining the context; no confirmation context meets this threshold.

Qwen3-14B confirms criterion-specific native responses: food selectivity is .282 [.199, .371] nats and service selectivity .404 [.294, .528]. Native signed determinant is .568 [.408, .751] nats² and normalized determinant .410 [.330, .493]. When every query position is prohibited from reading B, the A cache still formed from the full history yields .0289 [.0032, .0553] and .190 [.073, .310]. Thus criterion-related history can have distinct functional effects through the remaining computation.

The local label interface does not reproduce the full native responses. Its negative signed determinant also rules out describing every formation interface as a common-direction gain; the gated label interface has a small positive normalized response. The signed and normalized statistics weight contexts differently, so both are reported. Nonzero determinants reject the restricted functional common-direction account, not every nonlinear latent representation or an alternative retrieval policy.

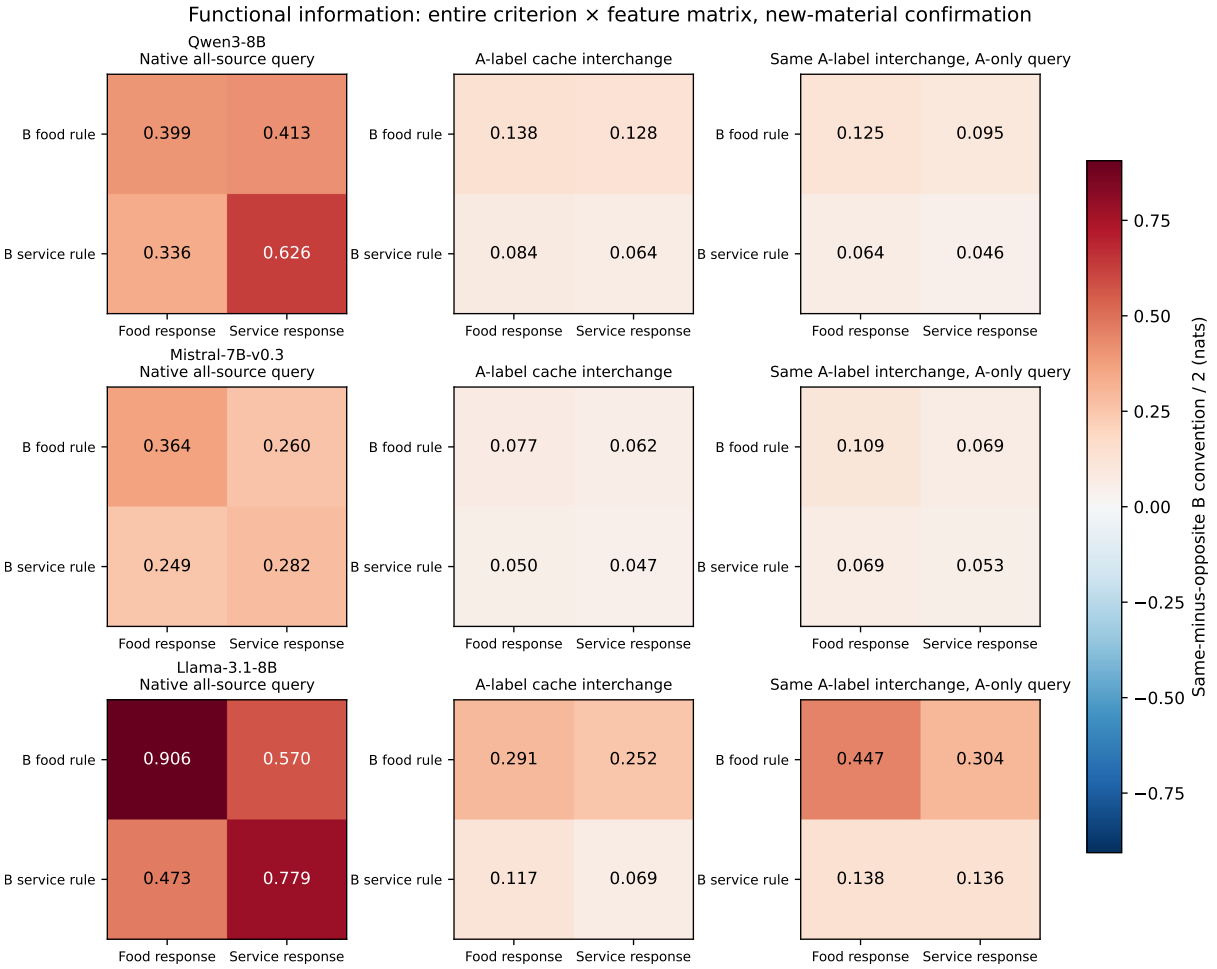


Figure 4: New-material confirmation, 32 contexts per family. Rows vary B's inferred criterion; columns measure the induced food/service response under independent B-convention reversal. The requested source remains A. Full matrices avoid equating one positive aspect response with complete criterion transfer.

Historical answer KV only partially predicts the actual criterion-label counterfactuals: native/gated raw correlations are .668/.665, RMSE .646/.878, compared with zero-prediction RMSE .746/1.018. This is a functional boundary of the formation reconstruction, not a reason to discard its successful namespace prediction. Local association transfer, whole-history criterion dependence, and source-conditioned use are different causal questions.

B Exact serialization, interchanges, and audit scope

Experiments use frozen Qwen3-8B and Qwen3-14B (both post-trained hybrid models), Mistral-7B-v0.3 base, and Llama-3.1-8B base, FP32 eager attention, PyTorch 2.8.0 and Transformers 4.57.6. The controlled sentiment task contains eight demonstrations from each source and eight new queries per context. Source names, private output direction, and demonstration order vary; the two private directions are balanced in confirmation. Shared outputs are `no/yes`; separate B outputs are `pass/flag`. A's outputs remain unchanged. Models process the serialized task as a continuation, without adding a chat template or generating reasoning. Qwen3-14B is a same-family size comparison; training data and recipes are not controlled between sizes. The two-aspect templates always mention food before service, so this experiment measures functional aspect distinctions under fixed order and does not separate semantic identity from position selection.

The actual serialized token stream is part of the method. Blocks, answer words, and separators are encoded separately. For Mistral, this introduces fixed extra spaces relative to literal whole-string encoding: the first native confirmation context has 351 rather than 335 tokens. Its B answer-message region includes the answer word plus three separator tokens per demonstration, totaling 32 sender

Qwen3-14B interface, n=32	Mean interaction J, nats [95% CI]	Signed determinant, nats² [95% CI]	Normalized determinant [95% CI]
Complete native histories	.343 [.265, .433]	.568 [.408, .751]	.410 [.330, .493]
Full history, A-only query	.058 [.029, .089]	.0289 [.0032, .0553]	.190 [.073, .310]
A-label interchange	−.0238 [−.0478, −.0025]	−.0201 [−.0374, −.0052]	.006 [−.121, .140]
A-label interchange, A-only query	.038 [.012, .068]	.0074 [−.0106, .0271]	.124 [.017, .242]
Historical answer KV at formation	.0060 [.0029, .0093]	.00068 [.00012, .00130]	.085 [−.042, .214]
Historical answer KV, A-only query	.0126 [.0076, .0188]	.00194 [.00031, .00474]	.072 [−.040, .190]
Separate-output label donor	−.0192 [−.0348, −.0060]	.00192 [−.00123, .00556]	.011 [−.145, .164]

Table 4: Complete supplementary readouts; Qwen3-14B interface, n=32.

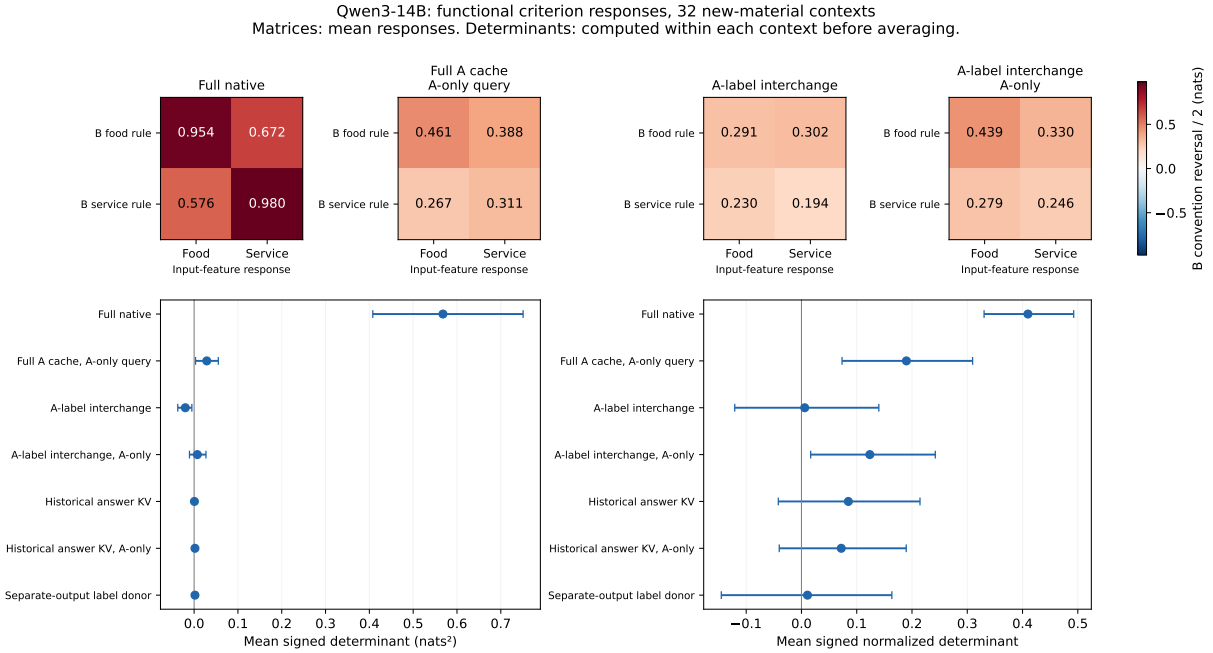


Figure 5: Qwen3-14B confirmation. Matrix means summarize functional responses; determinant intervals use determinants computed separately within each context. All seven predefined interfaces, including reverse and weak effects, are retained. Different cache interfaces do not establish different hidden algorithms.

positions; Qwen uses 16. Paired token positions and boundaries remain identical across the relevant manipulations. Native controls validate the actual stream, not a claimed equivalent whole-string encoding. Source code and independent tokenizer reconstruction preserve this disclosure.

Effects use 10,000 context-bootstrap resamples, seed 910; reconstruction intervals follow the frozen E109 analysis. Queries within a context are not treated as independent observations. Raw logits, full-vocabulary argmax, binary candidate-label accuracy, all conditions, and failed controls are retained. Confirmation uses new word/template pools; discovery-to-confirmation text overlap is zero for E113 and the Qwen E114 batches, but vocabulary/templates recur across contexts within a pool.

Instruction controls differ between experiment engines. E122/E123 add one sentence to native formation. In E124–E127, the stored `instruction` condition adds that sentence while also blocking private cross-source formation; it is a composite control, and its difference from native cannot be attributed solely to prompting. E124/E125 additionally include an instruction-free independent-formation reference. The primary formation, schema, and cache-composition contrasts do not use this instruction condition.

Self interchanges produce zero error. Cached versus full-token native query scoring differs by at most 4.96e−5 in Qwen and 7.63e−6 in Mistral confirmation. Cache fingerprints, layer consumption,

and matched token geometry are checked. An independent session reconstructs the token layouts and recomputes 559 numerical checks from raw records before reading author summaries, with maximum discrepancy 4.77e−15. Separate bootstrap resampling confirms the signed interactions; it is not represented as an independent GPU rerun.

The earlier E113/E114 package comprises five runs and 36,736 scores. The completed E115–E119 expansion comprises 24 discovery/confirmation runs, 576 contexts, 181,632 scores, and 2.393 recorded GPU-hours. Independent raw reviews recompute 36,750 numerical quantities (maximum discrepancy 1.13e−14), reconstruct actual token layouts, and check recorded all-layer consumption. This is independent analysis, not an independent GPU rerun. The E120/E121 native-operation and real-text additions comprise 16 runs, 384 contexts, 164,352 scores, and 2.540 recorded GPU-hours. Their independent raw audits recompute 18,064 scalars (maximum discrepancy 5.00e−15), reconstruct 2,112 actual token streams, and verify the recorded masks and reachable edges. No weights are trained. All model/seed/channel results remain available. Cards specify revisions, script hashes, and rebuilding commands; small summaries, figures, code, and audits are versioned, while raw JSONL and models remain at the documented asset paths.

C All formation/query policies and functional controls

Policy results average the two source requests, four B worlds, and eight queries within each context. Candidate accuracy and full-vocabulary argmax remain separate in raw records. `normal`, `answer_cross_off`, and `all_cross_off` use no trained parameters; the generated source ownership determines the mask. Both actors consume the same prefix cache.

All entries are accuracy percentages with context-bootstrap95%CI. Separate A/B, agreeing/disagreeing correct labels, all four criterion/convention worlds, single functions, one-sentence instructions, Food/Service/bias/FS responses, and all signed contrasts are in `results/e122/*_confirmation/analysis.json` and the independent raw audit. The single-aspect SST ability does not establish perfect dual-aspect primitive mastery.

Bars show the source-average primary; black circle/triangle show A/B means. These are not joint both-correct rates. Each paired contrast uses its context-level difference, not differences between separately constructed intervals.

Independent confirmation: each source's accuracy change, agreement/conflict of the correct labels, and alternative policies. All combined formation/query conditions and full-vocabulary scores are reported in the appendix.

D Native operation and real-text completeness

Paired signed B-rule-response differences in nats with95%CI. The operation acts at every prefix layer, masks strict historical senders, preserves self attention, and renormalizes the remaining attention. The restricted restoration opens only the historical answer→label edges; it does not restore original numerical messages or all native paths. E120 single within-answer-off controls, far-namespace effects, residuals, and signed/magnitude contrasts are in the card/results.

Each SST private direction has n16contexts; natural-text pool/split/source provenance is independently audited. The word-plus-separator sender regions differ by tokenizer. Llama fixed-word association signed effects are negative; all K/KV/V raw/rule reconstructions and zero predictions remain in the summaries.

Paired signed B-rule effects after operation removal and restricted restoration. The contrast in restoration direction shows why historical sharing cannot be described as uniformly harmful.

E Registered evidence map

F Real-text function policies and complete output readouts

Primary accuracy averages origin/cross worlds, two requested sources, and eight queries within each context. Within-class movement is reported separately. The original primary is candidate accuracy; full-vocabulary output and word-space diagnostics are separate secondary readouts. Shared-word candidate/full-vocabulary aggregates coincide in all six confirmation policies. Private polarity is balanced across contexts; source-specific conventions are inferred from examples, not provided as explicit rules.

These are paired context-bootstrap95%CI. The namespace contrast is not headroom matched; word

Model, confirmation n=32	Native	Answer formation	All cross formation	Query only	Answer + query	All formation + query
Qwen3-8B	71.58 [69.43, 73.44]	74.51 [72.66, 76.17]	62.60 [61.62, 63.67]	70.46 [67.87, 73.05]	73.78 [71.58, 75.73]	73.29 [70.80, 75.68]
Mistral-7B-v0.3	69.14 [67.63, 70.65]	72.56 [70.31, 74.85]	62.74 [60.99, 64.55]	70.85 [69.14, 72.46]	71.97 [69.87, 74.12]	73.54 [71.04, 75.88]
Llama-3.1-8B	73.83 [72.22, 75.44]	81.20 [79.10, 83.25]	68.55 [66.75, 70.26]	85.89 [84.08, 87.74]	86.77 [85.11, 88.53]	84.18 [81.30, 86.87]
Qwen3-14B	78.12 [75.68, 80.76]	80.91 [78.86, 82.86]	67.58 [66.02, 69.19]	84.23 [81.64, 86.82]	83.30 [80.81, 85.69]	82.91 [80.22, 85.64]

Table 5: Complete supplementary readouts; Model, confirmation n=32.

Model, n=32	Single A	Single B service/same	Single B service/opposite	Single B food/same	Single B food/opposite
Qwen3-8B	67.97 [64.06, 71.88]	83.59 [79.69, 87.50]	80.08 [76.17, 83.98]	70.70 [67.58, 73.44]	72.66 [69.14, 75.78]
Mistral-7B-v0.3	72.66 [69.53, 75.39]	72.27 [68.36, 75.78]	75.78 [72.27, 79.30]	76.95 [73.83, 80.08]	76.56 [73.44, 79.69]
Llama-3.1-8B	79.30 [75.00, 83.59]	87.11 [83.59, 90.62]	86.33 [82.03, 90.62]	82.42 [78.12, 86.72]	81.64 [76.95, 85.94]
Qwen3-14B	82.42 [78.91, 85.94]	85.55 [81.64, 89.45]	88.67 [84.77, 92.58]	83.59 [80.08, 87.11]	84.77 [80.86, 88.67]

Table 6: Complete supplementary readouts; Model, n=32.

meanings/priors also differ. No layer, head, intervention strength or context is selected to improve these results.

This classification uses the actual argmax token and the other source's two word IDs. The average other-source-word rates are 44.6–46.9%, but their A/B composition differs. In the shared-word case, surface words cannot identify their source: recording zero identifiable other-space words is not evidence of provenance isolation.

The private positive/inverted halves, actual full-vocabulary controls, all one-sentence instructions, complete A convention-response/signed/within movement readouts, and all 85 condition statistics remain in the experiment summaries and independent audit. Aggregate private functions are usable but not uniformly perfect; the distinct words also have asymmetric direction priors. Formation masks remove unequal sets of edges, renormalize attention, and allow states to evolve. Consequently the result identifies functional consequences of fixed policies, not isolated representation components or a tuned performance method.

The observer is the absolute context mean of the A-gold-oriented half contrast between B conventions, computed in A's fixed output space. It is not attention-based source ranking. Signed and matched within-class movement remain alongside it.

E123 freezes the scientific source at91374879 before the first score. Four checkpoints each execute discovery16 and confirmation32, totaling192contexts/130560scores. Scientific inputs, masks, full-output tokens, and small report data are independently checked; see E123 and its raw audit. The pinned SST downloader reproduces all three Arrow files and the Llama32 confirmation contexts byte-for-byte in the recorded environment, without modifying the frozen scientific modules.

G Shared identities and independent private supervision

E124 isolates both sources' private demonstration histories at every layer, retains original global positions, and places a public names-only list after the generic header but before all supervision. F controls demonstration access to the list; R controls every query position's direct access. All generic public tokens precede the list, preventing an indirect public relay when F is off. No output words, criterion or private convention are provided. True and unrelated lists have equal token length; numerical cache and full-output no-ops establish their complete invisibility under F0R0.

The primary is full-vocabulary accuracy, averaged uniformly over worlds, sources, and eight queries within each context. Identity-specific for-

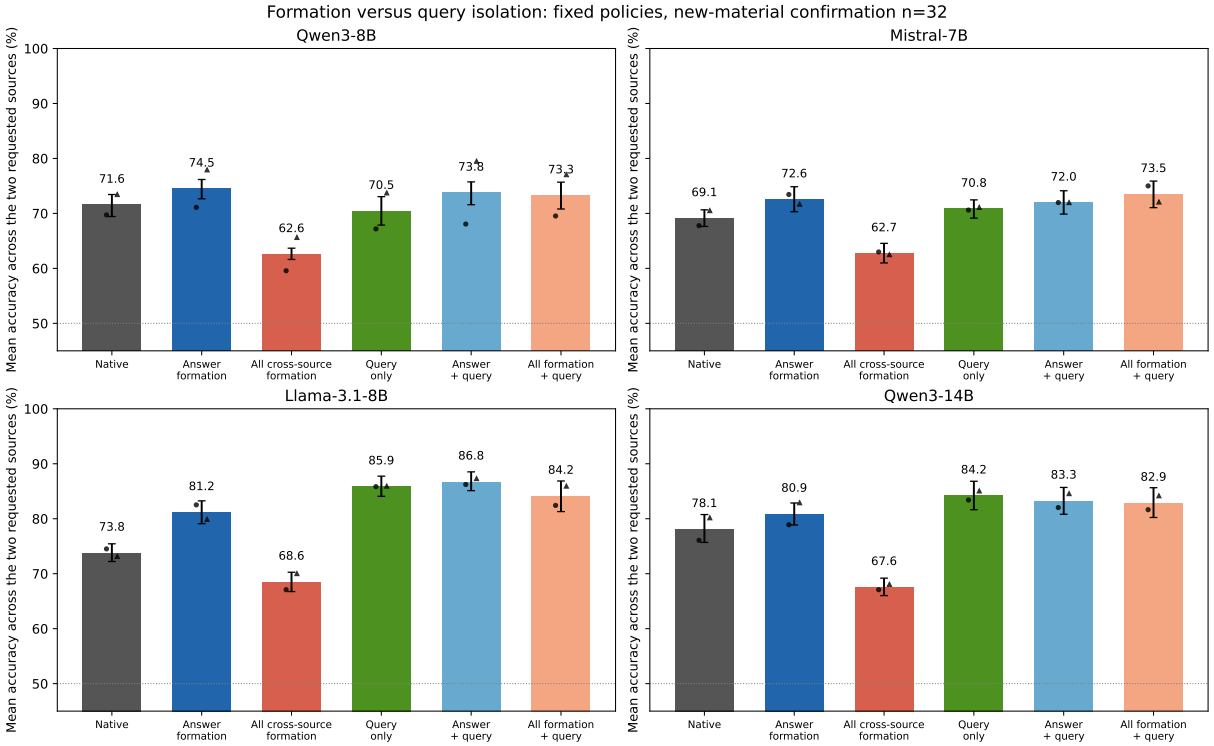


Figure 6: Complete policy accuracy

Model	Selective-minus-native A, pp	B, pp	Agreeing golds, pp	Disagreeing golds, pp	Selective-minus-all formation, pp
Qwen3-8B	1.37 [-0.68, 3.32]	4.49 [2.15, 6.84]	-2.34 [-4.00, -0.78]	8.20 [5.86, 10.45]	11.91 [10.25, 13.53]
Mistral-7B-v0.3	5.66 [2.54, 9.08]	1.17 [-1.27, 3.52]	-1.95 [-4.98, 0.88]	8.79 [6.15, 11.62]	9.81 [6.74, 13.04]
Llama-3.1-8B	8.01 [4.79, 11.33]	6.74 [4.49, 8.89]	-1.86 [-4.30, 0.39]	16.60 [13.87, 19.43]	12.65 [10.40, 15.04]
Qwen3-14B	2.83 [0.00, 5.47]	2.73 [0.10, 5.37]	-3.61 [-5.86, -1.56]	9.18 [6.15, 12.30]	13.33 [11.33, 15.38]

Table 7: Complete supplementary readouts; Model.

mation gain is [true F1R0−true F0R0]−[unrelated F1R0−unrelated F0R0]; reading gain substitutes F0R1. It is not a comparison to the shorter no-list prefix.

Qwen's formation identity gain is positive for A and B and both private directions. Mistral's true formation-only operation has opposite effects on A and B; Llama's positive candidate identity contrast does not transfer to the full-output primary. Mistral's reading identity contrast is positive but small. All 47 conditions, instructions, single-source functions, private directions, signed/absolute B-convention responses, and F×R contrasts are retained in the complete report and its summaries. There is no equivalence conclusion from a confidence interval crossing zero.

Three checkpoints each execute 16 discovery and 32 confirmation contexts, changing Alex/Sam to Robin/Jordan in confirmation. The 54,144 scores undergo independent numerical, token and graph review; 11,742 numerical comparisons match exactly. Whole-query mask consumption is checked in every layer. Independent review is not an independent GPU rerun. The names-only public context is not a full entity–attribute binding construction, and its partial/negative effects do not refute all binding mechanisms.

H Public output schemas and private rule composition

E125 keeps private cross-source demonstration access blocked at every layer. The public table precedes all supervision and specifies only source-to-word sets, not input–label pairs, criterion, or

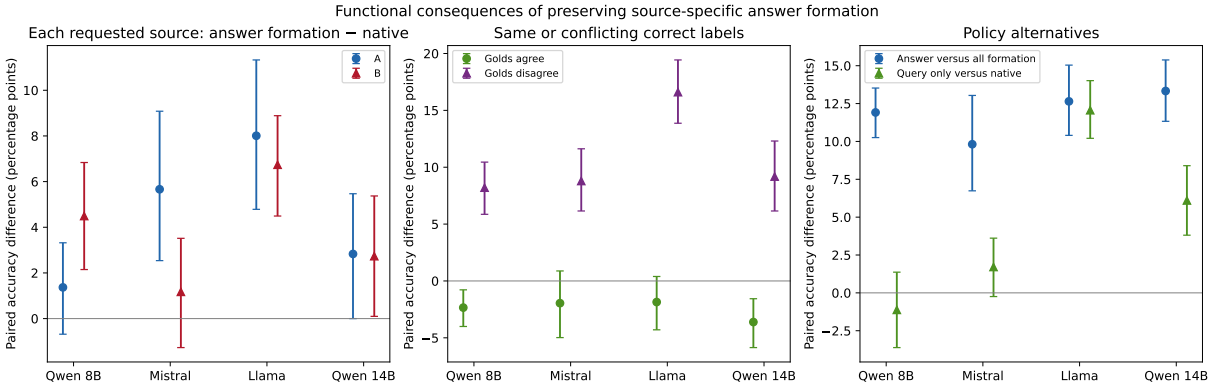


Figure 7: Functional source-policy effects and tradeoffs

Model, n=32	Full-source operation removal	A-only restricted restoration	Full-source restricted restoration
Qwen3-8B	1.649 [1.108, 2.250]	1.455 [0.999, 1.945]	-0.806 [-1.365, -0.266]
Mistral-7B-v0.3	1.054 [0.886, 1.227]	0.647 [0.463, 0.835]	0.140 [0.053, 0.231]
Llama-3.1-8B	0.808 [0.517, 1.109]	0.958 [0.635, 1.276]	-0.233 [-0.336, -0.122]
Qwen3-14B	1.329 [0.967, 1.717]	1.120 [0.673, 1.646]	0.039 [-0.217, 0.287]

Table 8: Complete supplementary readouts; Model, n=32.

private polarity. F controls all demonstration access; R controls direct access by every query token. Generic HEAD tokens precede the table, preventing a later public relay. True, unrelated-name, and swapped-relation tables have matched lengths and unchanged private token positions. Query reading otherwise sees both source caches. This is explicit relational supervision, not fully unsupervised coordination.

The primary difference-in-differences compares true and unrelated formation-only or reading-only gains. All three tables share identical relevant states and full-output argmax under F0R0. Below, confidence intervals are 10,000 context-bootstrap intervals; contexts, rather than individual queries, are the sampling units.

A correct schema helps more than its swapped counterpart, but the swapped schema also improves actual full-output accuracy and reduces other-source outputs in every model. Therefore a positive true−swapped contrast does not establish active wrong-space routing. Qwen's true formation operation costs 3.81 [1.86, 5.66] candidate-accuracy points; Mistral and Llama have uncertain A-specific identity gains. All private directions, A/B/world readouts, single functions, instructions, signed/absolute B-convention effects, and F×R interactions are retained in the complete report and per-run analysis files. A shared schema is not a lossless or universally sufficient composition method.

The frozen scientific source is `75f75665 82cbaf7cf80dd130b409b32436df281a 262884d8db381991437612b9` (preregistered commit a0a79918). Six runs include 144 contexts and 72,576 scores, recording 0.735239 GPU hours. The independent audit reconstructs 18,126 raw numerical entries with maximum error zero, actual tokenizer geometry, golds, 100,800 prefix and 302,400 query layer-mask calls, and 9,216 inert-table full-output comparisons. This is independent analysis rather than an independent GPU rerun. Fresh confirmation contexts and names do not constitute an entirely new benchmark.

I Equivalent-context cache composition

E126 changes only the order of two correct public source–word-set rows. Private examples, all public facts, source names, token lengths, and private global positions are unchanged. Private cross-source formation remains blocked at every layer, while every query position is denied direct schema access. Coherent pairs 00/11 combine whole source-owned K/V from one ordering; mixed pairs 01/10 combine one source from each ordering. Generic HEAD and unreadable schema cache states come from order0. Coherent11 is verified against the pristine order1 computation under the same query mask.

The primary is full-vocabulary accuracy, with equal weights over worlds, sources, and queries

Model, n=32	Single-source positive / inverted accuracy, %	Native A-formation magnitude, nats [95% CI]	KV rule r / RMSE / zero RMSE
Qwen3-8B	98.4 / 88.3	2.151 [1.487, 2.895]	0.931 / 1.363 / 3.658
Mistral-7B-v0.3	99.2 / 96.9	1.091 [0.925, 1.258]	0.948 / 0.336 / 1.048
Llama-3.1-8B	97.7 / 99.2	0.689 [0.488, 0.901]	0.930 / 0.309 / 0.840
Qwen3-14B	99.2 / 93.8	1.678 [1.340, 2.052]	0.966 / 0.519 / 1.949

Table 9: Complete supplementary readouts; Model, n=32.

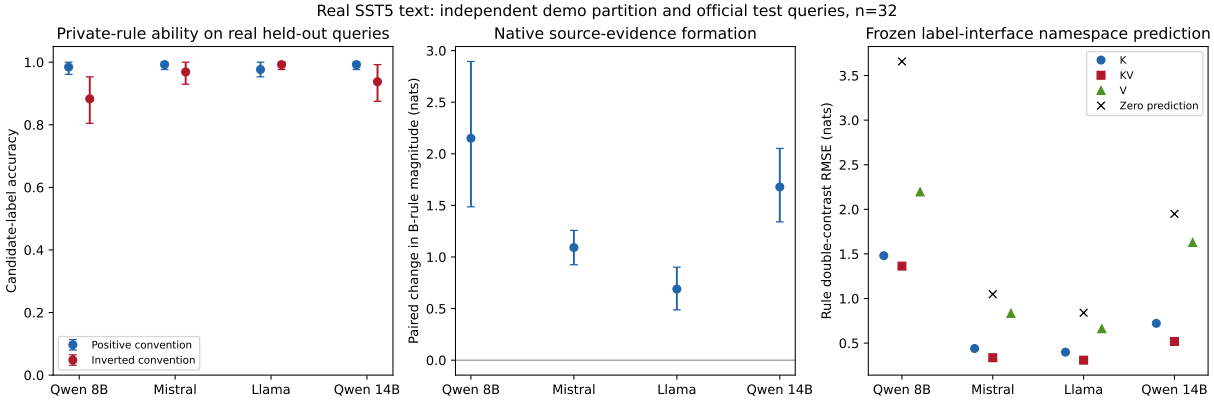


Figure 8: Real-text formation and private functions

within a context. The paired coherent−mixed contrast is .5[(00+11)−(01+10)]. It cancels additive single-cache quality effects but can still reflect nonlinear output scoring, attention competition, or binding. It does not by itself identify nonlinear latent representations. Requested-source-only reading makes the balanced contrast zero by the combination design; individual function scores must be reported separately.

Both public orders have strong source-only function scores, but they are not functionally identical. Joint order11−00 is negative in Qwen and Mistral. Their primary correct-margin point estimates are also negative with intervals spanning zero; improved full-output use is not a uniform positive shift in every score. Mistral's A-specific joint full accuracy is only 42.58% in00 and 33.40% in11, despite strong requested-source-only scores. All actor, world, private direction, margin, convention-response, single-function, and one-instruction results are retained in the complete report.

Six runs comprise144contexts/49536scores and0.627887GPU hours. Source `ad2c6f7af7ab81966ea1253a80e7006747f3485b182fddd1453ffbb1ceb45975` was frozen at377de6c4 before scoring. The independent audit reconstructs12462 numerical entries with maximum error0,1584 actual token/mask layouts,49536 golds,52800 prefix and206400 query layer-mask calls,9216 mixed-own argmax controls, and4608 A own-world comparisons. No independent GPU rerun is claimed. The portable audit confirms Llama32 material bytes from the pinned public dataset.

I.1 Post-hoc selected-output compatibility

After the complete E126 confirmation, we classify actual argmax outputs in the opposite-convention world against both sources' private gold. The same-convention world cannot distinguish the two mappings and is excluded from this diagnostic. Every context and both requested sources remain included; the unit is still context. This is an additional observation, not a change to the frozen primary or a new model run.

These categories use the task's canonical code indices rather than the words' natural semantic equivalence. They show compatibility of an observed output with a demonstrated mapping, without claiming to observe an internal called rule. The predominant change is in outputs compatible with the other source's convention. Own-space wrong-code outputs also increase in Qwen by 2.54 [0.68, 4.88] points and Llama by 0.68 [0.10, 1.27], so this is not a uniform reduction of every error type. These opposite-world categories are not a mediated fraction of the both-world primary. Full pair-specific rates and discovery results are in the post-hoc report and raw recomputation.

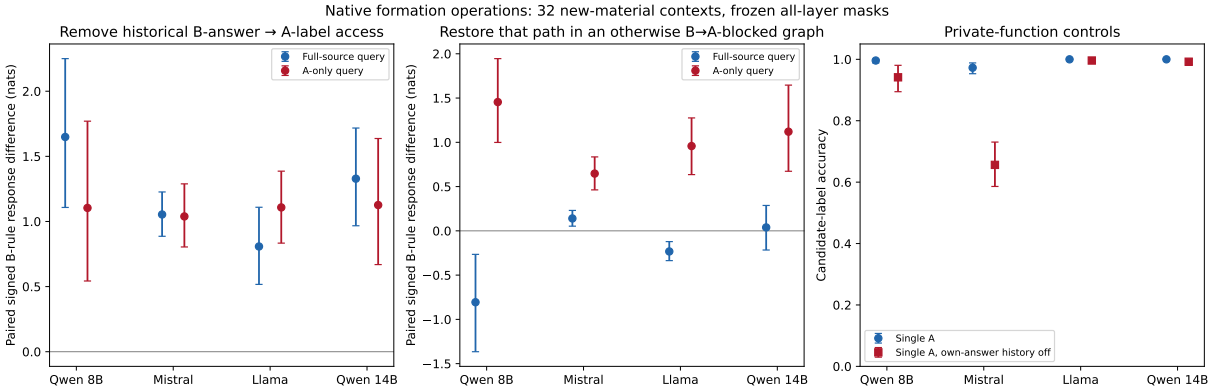


Figure 9: Native operation and restricted restoration

Manuscript claim	Records
Positional access does not isolate relation dependence	E90, E91
Output-dependent formation reconstruction, C21	E109, E111, E112, first independent audit
Fixed-label partial association content, C22	E111, E113, E114
Native operation contribution and restricted restoration	E120, independent operation audit
Reliable private functions and formation on natural text	E121, independent data and mechanism audit
Native conditional effects and interaction, C23	E114, new independent audit
Functional criterion response and its formation-interface boundary, C24	E117, E118, E119, E118 audit, E119 audit
Llama, larger checkpoint, and real-text coverage	E115, E116, E119; wave audit
Limits and failed functional interfaces	E50–E52, E99, E107, E110; full ledger in CLAIMS
Joint source-function formation and query policies	E122, independent raw audit

Table 10: Complete supplementary readouts; Manuscript claim.

J Shared-output cache compatibility and source-conditioned mapping use

E127 holds the original comments, source identities, class-index conventions, and global private positions fixed across shared and separate namespaces. Two orders of the same correct public facts form whole source-owned K/V caches. Every private cross-source formation edge is blocked; every query position is denied schema access. The 00/11 pair uses one common ordering and 01/10 mixes orderings. Each cache is scored under joint and requested-source-only reading. Both private directions, sources, and all predefined worlds remain included.

The preregistered primary is **shared outputs, opposite private conventions, joint coherent-minus-mixed full-vocabulary accuracy**. Other worlds and separate word sets are secondary. Shared words cannot identify which source an answer came from: the stored `other_source_word=False` is unidentifiability, not zero leakage. Full argmax, candidate choice, and raw label logits are retained. Neither the source-average primary nor the local functions measure both answers being simultaneously correct.

J.1 Functional request differences and common shifts

For shared outputs in the opposite-convention world, define $S = \frac{1}{2}\mathbb{E}_q[g_A(q)(z_A(q) - z_B(q))]$ and $C = \frac{1}{2}\mathbb{E}_q[g_A(q)(z_A(q) + z_B(q))]$. Absolute common magnitude averages the absolute value per query before averaging. S is also the mean correct margin over both requested sources, because their gold labels are opposite. Its change excludes a single request-independent logit offset as the complete explanation; it does not identify a hidden binding representation or a unique retrieval algorithm. Separate outputs instead use the logsumexp difference between word sets, so their functional units must not be treated as the same shared-word measure.

J.2 Local functions and incomplete joint use

Strong private functions coexist with incomplete joint selection. Llama's primary is small, its B-specific interval spans zero, and the positive effect is not uniform across private directions. Mistral's two coherent orders differ. The balanced own-only linear coherence contrast is zero by construction, whereas absolute-common observers may have nonzero contrasts because of their nonlinear def-

Model / output words, n=32	Native candidate / full, %	Selective candidate / full	All formation off candidate / full	Query-only candidate / full	Answer + query candidate / full	All formation + query candidate / full
Qwen3-8B / shared	84.86 / 84.86	92.19 / 92.19	75.00 / 75.00	84.47 / 84.47	93.85 / 93.85	94.34 / 94.34
Qwen3-8B / far	94.24 / 92.09	94.34 / 92.09	94.34 / 51.46	94.34 / 94.34	94.63 / 94.63	93.16 / 93.16
Mistral-7B-v0.3 / shared	88.77 / 88.77	94.14 / 94.14	73.54 / 73.54	94.73 / 94.73	96.39 / 96.39	93.16 / 93.16
Mistral-7B-v0.3 / far	98.05 / 96.00	98.05 / 94.73	95.21 / 53.71	97.27 / 97.27	97.46 / 97.46	95.61 / 95.61
Llama-3.1-8B / shared	81.25 / 81.25	90.72 / 90.72	75.20 / 75.20	97.95 / 97.95	99.12 / 99.12	99.12 / 99.12
Llama-3.1-8B / far	97.95 / 95.41	98.54 / 95.12	94.82 / 53.81	96.58 / 96.58	96.97 / 96.97	95.31 / 95.31
Qwen3-14B / shared	89.94 / 89.94	97.36 / 97.36	75.00 / 75.00	93.75 / 93.75	97.27 / 97.27	98.05 / 98.05
Qwen3-14B / far	95.90 / 95.80	97.27 / 97.07	96.68 / 52.83	92.97 / 92.97	94.63 / 94.63	96.68 / 96.68

Table 11: Complete supplementary readouts; Model / output words, n=32.

Model	Selective shared gain A, pp	B, pp	Opposite conventions, pp	Same conventions, pp	Shared-minus-separate gain, pp
Qwen3-8B	4.49 [1.56, 7.03]	10.16 [6.05, 14.26]	15.23 [11.91, 18.75]	-0.59 [-2.15, 0.98]	7.23 [5.08, 9.47]
Mistral-7B-v0.3	4.49 [0.98, 8.20]	6.25 [3.32, 9.38]	12.11 [8.20, 16.21]	-1.37 [-3.71, 0.20]	5.37 [2.73, 7.81]
Llama-3.1-8B	7.42 [4.69, 10.16]	11.52 [8.20, 14.65]	19.14 [15.04, 23.24]	-0.20 [-0.59, 0.00]	8.89 [6.84, 10.94]
Qwen3-14B	4.88 [2.15, 8.20]	9.96 [6.05, 14.06]	15.23 [11.52, 19.14]	-0.39 [-1.37, 0.59]	6.05 [4.30, 7.91]

Table 12: Complete supplementary readouts; Model.

inition. None of these identities demonstrates a separate latent module. All sources, worlds, private directions, single functions, native and instruction-plus-isolation controls, candidate/full scores, B-code response, and raw common shifts appear in the complete report.

Eight fixed runs contain 192 contexts and 132,096 scientific scores. The completed-run timers total 1.950864 GPU hours; they exclude preserved partial-run computation, the zero-score load failure, startup reservations, and model transfer. Scientific source `cef7952d4ef556ab ad45c9c2e6d6f351e3b0a1b3fffa9232 b9eb728806c943de` was frozen before scoring at commit4b8ee609. The independent CPU record covers all 192 contexts and actual tokenizer/configuration layouts; the complete raw audit reports numerical recomputation, golds, masks, and actual per-layer consumption. These are independent analyses, not independent GPU reruns.

Qwen3-14B's first confirmation load failed before any scientific score after an external job occupied its GPU. The pinned checkpoint was then verified on another empty node. The initial discovery run's ten complete contexts and raw records remain preserved separately. Identical contexts were rerun in the full fixed discovery batch, without seed or outcome selection: all 6,880 matching argmax decisions and candidate signs agree across hardware; maximum absolute logit-difference and individual label-logit deviations are .001028 and .002842. These hardware comparisons are separate from the same-run FP32 controls and are not additional samples. Checkpoint identity and the operational records preserve the migration. Portable reconstruction reproduces the 32 Llama confirmation contexts byte-for-byte from the pinned public SST5 files.

Coherent-minus-mixed cache effects on held-out contexts. Left: identical words and conflicting conventions, the frozen primary. Right: separate words and both convention worlds, a secondary reference. The panels use different world scopes.

Model	Far/all-formation-off full accuracy A, %	B, %	Other-source word output A, %	B, %
Qwen3-8B	48.24 [40.43, 56.05]	54.69 [45.70, 63.48]	51.17 [43.36, 58.98]	42.58 [33.40, 51.76]
Mistral-7B-v0.3	30.86 [21.68, 40.62]	76.56 [68.75, 83.98]	66.99 [56.84, 76.76]	22.27 [14.84, 30.08]
Llama-3.1-8B	64.65 [58.01, 71.29]	42.97 [34.96, 50.98]	34.96 [28.32, 41.60]	55.27 [47.46, 63.28]
Qwen3-14B	46.09 [39.65, 52.34]	59.57 [53.52, 66.02]	53.71 [47.46, 60.16]	38.48 [32.03, 44.73]

Table 13: Complete supplementary readouts; Model.

Model	Single A, %	B opposite shared	B same shared	B opposite far	B same far
Qwen3-8B	95.31 [92.58, 97.66]	94.14 [90.62, 97.27]	97.27 [94.92, 99.22]	92.97 [88.28, 96.88]	94.14 [90.23, 97.66]
Mistral-7B-v0.3	98.83 [97.27, 100.00]	98.05 [94.92, 100.00]	98.44 [96.88, 99.61]	98.83 [97.66, 100.00]	98.05 [96.48, 99.61]
Llama-3.1-8B	98.44 [96.88, 99.61]	99.22 [98.05, 100.00]	99.61 [98.83, 100.00]	94.14 [89.84, 97.66]	89.06 [83.58, 94.14]
Qwen3-14B	98.05 [96.48, 99.61]	96.88 [93.75, 99.22]	98.44 [96.48, 100.00]	95.70 [93.75, 97.66]	96.48 [92.97, 99.22]

Table 14: Complete supplementary readouts; Model.

K Compatibility when input criteria differ

E128 uses controlled restaurant comments with two independently varied aspects and both private code directions. A follows Food; B has four criterion×code worlds. Both sources share no/yes, public facts and private global positions match across orders, private cross-source formation is blocked, and the whole query cannot read the public table. The frozen primary is the origin world (A Food/B Service, same code), the four aspect-conflicting queries, averaged over both requested sources. This differs from E127's convention primary and does not inherit its stronger private-function scores.

For origin, let f and s denote the Food and Service signs oriented by A's private code. We report $T=\frac{1}{2}$ E[(zA−zB)(f−s)] over all eight balanced queries. This equals the average correct margin of the two sources on the four conflicting queries: it is an equivalent output projection, not an independent probe or evidence of input filtering. The complete Food/Service/bias/product basis and common shifts remain available.

K.1 Actual function scores on the primary domain

The single-function conflicting-query values are an explicitly **post-hoc descriptive scope audit** of already predefined controls, recorded by the independent reviewer. They neither change the primary nor add model runs. All-query controls in the full report have a different domain and cannot establish capability on the primary conflict cases. Qwen3-8B's Food is weak; Mistral's Food score is modest and its Service interval includes chance; Qwen3-14B's matched Food/Service functions are above chance. Its small positive compatibility effect therefore coexists with a substantial private-to-joint gap. This prevents interpreting the weaker effects as absence of criterion information, or the positive margin as successful criterion composition.

All 85 interfaces, four worlds, private directions, source-specific intervals, both coherent orders, native and **pure one-sentence instruction** conditions, full/candidate outputs, and fixed-word response bases are preserved in the complete report and independent disclosure. Aspect order remains fixed, and whole-cache interchanges remain compatible with contextual binding, calibration, and criterion-conditioned retrieval.

Eight runs comprise192contexts/130560scores and1.237902 recorded GPU hours. Scientific source `f0a0146708e193e017eb33a1caf955d8e4d4bf22d91d73abb403b500be92265a` was frozen at5e9549ed before scoring. The independent raw audit reconstructs38760 numerical entries with maximum error0 and checks actual masks, golds, full argmax, cache origin, and the margin identity. The author report check adds38764 numerical entries and352 table rows; these comparison counts overlap the same observations and do not add experimental sample size. No independent GPU rerun is claimed.

All four cache pairs on the frozen criterion-conflict domain, confirmation n32 per checkpoint.

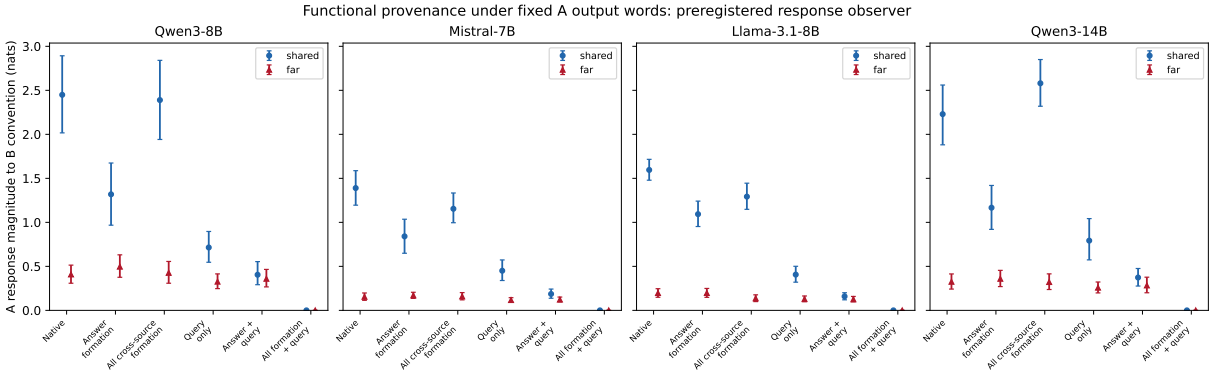


Figure 10: B convention response on fixed A words

Model, n32	Formation identity gain, pp	Reading identity gain, pp	true F0R0 full/candidate, %	true F1R0 full/candidate, %
Qwen3-8B	4.59 [2.73, 6.64]	0.29 [-0.78, 1.46]	52.73/93.16	58.98/93.65
Mistral-7B-v0.3	-1.07 [-2.05, -0.20]	0.49 [0.10, 0.98]	53.32/91.41	52.83/92.97
Llama-3.1-8B	-0.68 [-1.95, 0.49]	-0.20 [-0.88, 0.39]	53.22/93.95	54.59/93.95

Table 15: Complete supplementary readouts; Model, n32.

Scores average separate source requests; they do not measure simultaneous two-source success.

L Does a pre-input source block deploy a source-specific criterion?

E129 retains the controlled Food/Service material and crosses B's criterion with its private code while fixing names, B label streams, input multisets, and global positions. Private demonstration histories form independently, without a public schema. Before any test comment, the complete `Annotator: name / Comment:` block reads both private histories; the cue stage does not use an oracle source mask. We exchange only this entire multilayer cue K/V block, preserving every recipient demonstration cache. The primary body graph then reads recipient B, HEAD, and the cue; joint-body reading is secondary. Two same-code recipients, four donor worlds, and CueA/CueB controls remain fixed.

Let $\delta t = \frac{1}{2}(z_{cue}(t,+) - z_{cue}(t,-))$. Its projections on private-code-oriented Food and Service signs form a two-row functional matrix M. The two co-primary quantities are det(MCueB) and det(MCueB−MCueA), calculated separately for every recipient/context before averaging the two recipients and bootstrapping contexts. A positive difference determinant alone can result from subtracting two rank-one matrices, so the first co-primary and correct row selectivities cannot be omitted. These are output functions, not measured latent dimensionalities.

Both confirmation co-primary intervals span zero. Llama's small positive Source-specific determinant in discovery does not confirm. Qwen3-14B retains small criterion-related responses, including a mean row-selectivity interaction of0.00160 [0.00024,0.00300]nats. This does not establish correct deployment of both criteria; it also prevents claiming that the cue contains no task information.

The matched single functions are usable, while transplantation does not establish a complete criterion×code function. Every patch reports accuracy against both recipient and donor rules, so following the donor is not mistakenly judged only by recipient accuracy. The full report preserves all64 conditions, two body graphs, source cues, recipients, worlds, private directions, four response bases, normalized rotations, actual output tokens, and function controls.

Native early/late uses the same formation and reading graphs, changing only field order. Independent early prepares a cue with access to both histories and then uses an own-only body; independent late uses own-only reading throughout, so their difference also changes the graph. The pure one-sentence instruction retains native formation. In Qwen3-14B origin B it gives84.38% candidate accuracy but44.14% full accuracy, with55.08% of outputs in the label set; the actual-output cost is retained rather than described as recovery.

This is a test of cues prepared from **independently formed private histories**, not a negative

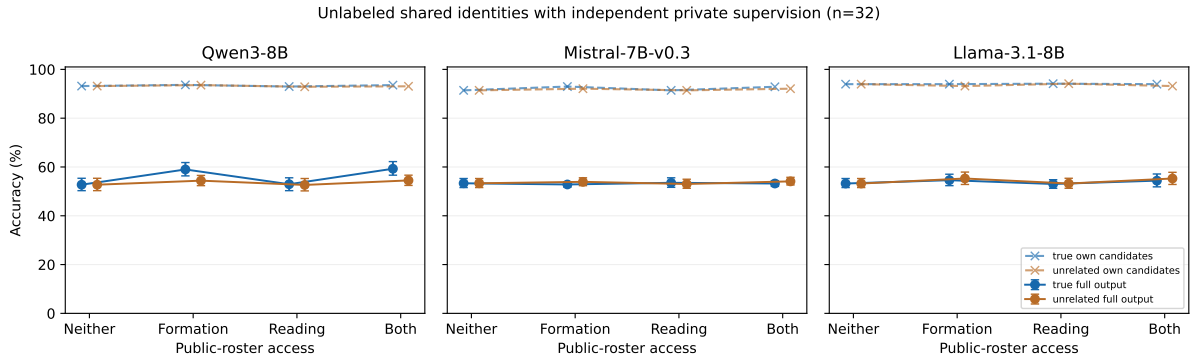


Figure 11: Shared identities at formation and reading

Model / phase	Formation true−unrelated, pp	Reading true−unrelated, pp	Formation true−swapped, pp	Swapped formation gain, pp
Qwen3-8B / discovery	20.31 [16.99, 23.83]	0.39 [-0.39, 1.17]	10.16 [7.03, 12.89]	9.96 [6.25, 13.87]
Mistral-7B-v0.3 / discovery	10.94 [6.25, 16.60]	0.20 [-0.98, 1.37]	6.05 [2.34, 10.16]	6.64 [3.71, 10.35]
Llama-3.1-8B / discovery	8.98 [5.08, 12.70]	0.00 [-0.59, 0.59]	12.50 [8.79, 16.80]	0.20 [-2.54, 2.54]
Qwen3-8B / confirmation	26.66 [23.05, 30.47]	0.78 [-0.29, 1.95]	11.72 [7.91, 15.72]	16.99 [13.77, 20.31]
Mistral-7B-v0.3 / confirmation	7.71 [5.08, 10.45]	0.10 [-0.29, 0.49]	4.39 [2.15, 6.74]	6.05 [4.10, 8.11]
Llama-3.1-8B / confirmation	8.30 [6.05, 10.84]	0.29 [-0.20, 0.78]	5.47 [2.44, 8.59]	2.64 [0.10, 5.08]

Table 16: Complete supplementary readouts; Model / phase.

result about every native task state. It remains compatible with criterion-conditioned retrieval, task information in other positions, and information becoming deployable after query input. CueA/B differs in lexical identity; donor-world subtraction cancels only additive, world-invariant name effects. General task-state transfer is already established by Hendel et al., just-in-time task representations (Li et al., 2025), and representative task-vector analysis. This interface test neither claims a first task vector nor retracts the demonstrated historical formation mechanism.

Four runs comprise96contexts/49152scores and0.413184 recorded GPU hours. Source `a177434cf425889df1fcc8b3607e6d50 57a26080ad467e7999d002e6504b00bd` was frozen at57ca241a before scoring. The independent raw audit checks20580 numerical quantities, both gold definitions, actual masks and58752 prefix/27648 cue/221184 body layer-mask calls. Across all four runs, the maximum full-token/cached difference is5.5313e−5nats; graph-zero and self-interchange checks are0. These are independent analyses, not independent GPU reruns.

Complete confirmation functional matrices, own-body scope. F/S denote Food/Service; arrows label the donor criterion row and query feature projection. Intervals use contexts rather than individual queries.

L.1 E130: Source-cue transfer under different formation policies

E130 follows the E129 cue/body/readout design and holds the recipient history fixed within each newly sampled context, while preparing donors under three historical policies: native contextualization, selective removal of foreign-answer→own-label edges, and complete private-history isolation. All donors are transplanted into the same independently formed recipient caches. The two fixed co-primary determinants use native donors; every policy contrast and actual donor/recipient function is reported.

All four intervals span zero. Qwen3-14B nevertheless retains a native CueB Food row selectivity of .02144 [.00972,.03348] nats; its Service selectivity is −.01465 [−.02661,−.00353] nats. The response therefore cannot be summarized as no cri-

Model, confirmation n32	True F0R0 full / candidates, %	True F1R0 full / candidates, %	Formation identity gain A / B, pp	Swapped other-word change, pp
Qwen3-8B	51.27/95.12	79.98/91.31	29.88 [23.05, 36.72] / 23.44 [17.97, 29.69]	-18.75 [-21.78, -15.92]
Mistral-7B-v0.3	50.00/91.02	60.45/95.70	4.49 [-0.20, 9.38] / 10.94 [7.42, 14.45]	-4.79 [-6.74, -2.93]
Llama-3.1-8B	53.22/94.24	61.33/95.12	0.20 [-5.27, 5.66] / 16.41 [10.35, 23.05]	-3.22 [-5.47, -0.98]

Table 17: Complete supplementary readouts; Model, confirmation n32.

Model / phase	Joint coherence, pp	Candidate coherence, pp	Joint order11−00, pp	Coherence A / B, pp
Qwen3-8B / discovery	19.43 [15.82, 23.24]	1.07 [0.00, 2.15]	0.20 [-5.27, 5.47]	20.12 [13.28, 27.73] / 18.75 [14.26, 23.83]
Mistral-7B-v0.3 / discovery	15.33 [11.33, 19.53]	0.29 [0.00, 0.59]	4.88 [-2.73, 12.89]	25.39 [18.55, 33.20] / 5.27 [2.54, 8.20]
Llama-3.1-8B / discovery	7.91 [5.08, 10.84]	0.10 [-0.49, 0.68]	-0.98 [-4.88, 2.34]	9.18 [4.49, 14.65] / 6.64 [3.32, 10.16]
Qwen3-8B / confirmation	23.49 [20.80, 26.07]	1.12 [0.24, 2.05]	-4.10 [-7.62, -0.59]	21.19 [17.58, 24.90] / 25.78 [21.09, 30.47]
Mistral-7B-v0.3 / confirmation	10.40 [7.96, 13.04]	0.63 [0.29, 0.98]	-5.27 [-10.25, -0.49]	17.48 [13.18, 22.07] / 3.32 [1.46, 5.37]
Llama-3.1-8B / confirmation	5.42 [4.00, 6.79]	0.20 [-0.24, 0.63]	1.46 [-0.98, 3.91]	6.05 [4.20, 8.11] / 4.79 [2.73, 6.74]

Table 18: Complete supplementary readouts; Model / phase.

terion information, or as correct transfer of both criteria. Changing formation policy did not establish the complete pre-input function at this interface. The full report retains all 136 interfaces, policy contrasts, private directions and recipient/donor full-output functions. This follow-up tests the source-cue interface; the historical formation, relation-content and native-operation results remain separate evidence.

The four fixed runs contain 96 contexts and 104448 scores, with .805577 recorded single-GPU run hours. Source 7ef59b3f… was frozen at fa41da09 before scoring. The independent audit verifies 81188 raw-derived numerical quantities and all actual prefix/cue/body mask calls; full-token/cached error is at most 2.48e−5 nats. The final report check verifies 28728 displayed numbers and retains the disclosed Llama discovery stdout reading-order exception. No independent GPU rerun is claimed.

M Matched cross-size formation, operation, and source-function tests

E131 preserves the original E121 formation and reconstruction engine, E120 native-operation engine, and E123 real-text function-policy engine. The preregistered matrix contains Qwen3-32B and Llama-3.1-70B discovery n16→confirmation n32, and fresh same-family 8B n32 anchors for all three stages. At this snapshot, all six 8B stages and all six 32B stages are complete and independently checked; the six 70B stages await completed scientific batches. Uncompleted runs supply no scientific results here.

Contexts are matched byte-for-byte **within each family and stage**. Formation and function use separately sampled SST5 contexts; native operation uses controlled restaurant contexts. Confirmation seeds are 131102/131101/131103 for these stages. Parent analyses and gold weighting remain unchanged, and comparisons do not pool different stages or treat a recipe change as an isolated capacity intervention. Each large model uses all its layers with frozen FP32 weights, eager attention, no CPU offload or quantization; Qwen3-32B uses two GPUs. Model pins, dataset revisions, actual consumed masks, and engine/parent AST checks are recorded in the E131 manifests.

These are signed B-rule contrasts, not absolute magnitudes. The restoration direction agrees across these checkpoints while the A-only restoration magnitude differs (large−small −1.368 [−2.299, −.474] nats). The 32B formation-stage signed interaction interval contains zero (−.037 [−.525, .462]), while the absolute interaction is positive (.953 [.162, 1.749]); cancellation of signed con-

Model, confirmation n32	Own-only00 / 11, %	Own-only01 / 10, %	Joint other-word coherence, pp
Qwen3-8B	92.09 / 93.36	91.99 / 93.46	-25.88 [-28.66, -22.90]
Mistral-7B-v0.3	96.68 / 96.78	96.39 / 97.07	-10.64 [-13.33, -8.11]
Llama-3.1-8B	93.16 / 92.58	92.77 / 92.97	-5.76 [-7.18, -4.30]

Table 19: Complete supplementary readouts; Model, confirmation n32.

Confirmation model	Coherent-minus-mixed other-source rule-compatible outputs, pp [95% CI]	Other-source outputs matching requested canonical code, pp [95% CI]
Qwen3-8B	−23.73 [−27.44, −20.02]	−1.37 [−2.25, −0.59]
Mistral-7B-v0.3	−10.06 [−12.79, −7.42]	−0.39 [−1.07, 0.20]
Llama-3.1-8B	−4.88 [−6.64, −3.22]	−0.59 [−1.17, −0.10]

Table 20: Complete supplementary readouts; Confirmation model.

text effects does not remove the separately measured formation contribution. All twelve channel/target/background reconstructions remain reported, including partial residuals.

The 32B real-text function primary averages both requested sources, origin/cross convention worlds, and queries at the context level. Full-vocabulary native accuracy is 87.99 [86.23, 89.75]%, selective formation 95.70 [93.55, 97.36]%, and complete cross-source formation removal 75.29 [74.12, 76.27]%. The selective-minus-native gain is 7.71 [5.76, 9.57] points; A gains 7.42 and B 8.01 points. Opposite-convention gains are 15.82 points, whereas the same-convention contrast is −.39 with an interval containing zero. Selective mean accuracy is not joint correctness: the latter is 92.19%. Query-only isolation improves the mean by 5.86 [4.20, 7.52] points.

Separate-word complete isolation retains 95.31% candidate accuracy but has 52.64% full-output accuracy and 9.96% joint correctness. Thus larger size does not make within-word-space accuracy a sufficient measure of using both source functions. All directions, private controls, instructions, other-source outputs, discovery results, and 36 fresh paired contrasts are linked in the complete E131 report, raw-first audit, and each stage's analysis/full-output summary. No coefficients or conditions are chosen from the cross-size results.

A first TF32-accelerated implementation failed the existing cached/full .001-nat tolerance before any scientific row; those empty failed attempts and their resource use remain archived. Restoring full FP32 GEMM preserved all materials, operators, and analyses. Valid completed stages have errors at most 6.68e−5 nats with self/graph checks zero. Independent analysis reproduces the original raw readouts and actual graph/gold layouts; it is not a second GPU execution.

Model / phase	Primary, pp [95% CI]	Shared/all worlds, pp	Separate/all worlds, pp
Qwen3-8B / discovery	6.84 [4.10, 9.77]	3.61 [2.25, 5.08]	17.09 [14.26, 19.63]
Mistral-7B-v0.3 / discovery	5.47 [3.71, 7.81]	2.64 [1.76, 3.81]	12.60 [8.50, 16.99]
Llama-3.1-8B / discovery	0.78 [0.00, 1.76]	0.39 [0.00, 0.88]	5.47 [2.83, 8.01]
Qwen3-14B / discovery	2.54 [0.98, 4.10]	1.46 [0.59, 2.25]	20.51 [16.89, 24.71]
Qwen3-8B / confirmation	8.89 [6.64, 11.13]	4.64 [3.52, 5.71]	23.49 [21.29, 25.73]
Mistral-7B-v0.3 / confirmation	5.86 [4.10, 7.71]	2.98 [2.10, 3.91]	9.13 [6.84, 11.43]
Llama-3.1-8B / confirmation	0.78 [0.10, 1.56]	0.39 [0.05, 0.78]	6.49 [4.93, 8.06]
Qwen3-14B / confirmation	9.38 [7.32, 11.52]	4.64 [3.56, 5.76]	21.44 [18.75, 24.02]

Table 21: Complete supplementary readouts; Model / phase.

Confirmation model	Shared S coherence, nats	Signed common coherence	Absolute common coherence
Qwen3-8B	0.63 [0.53, 0.74]	0.00 [-0.06, 0.06]	-0.05 [-0.10, -0.01]
Mistral-7B-v0.3	0.13 [0.11, 0.16]	0.04 [0.01, 0.07]	0.00 [-0.02, 0.02]
Llama-3.1-8B	0.03 [0.02, 0.03]	0.00 [-0.00, 0.01]	0.00 [-0.00, 0.01]
Qwen3-14B	0.50 [0.44, 0.56]	-0.00 [-0.07, 0.07]	-0.02 [-0.07, 0.03]

Table 22: Complete supplementary readouts; Confirmation model.

Confirmation model	Shared/opposite joint00 / 11, %	Shared/opposite own-only00 / 11, %	Joint order11−00, pp
Qwen3-8B	63.87 / 64.84	90.82 / 91.41	0.98 [-2.34, 4.10]
Mistral-7B-v0.3	59.18 / 56.25	96.88 / 96.09	-2.93 [-5.08, -0.78]
Llama-3.1-8B	53.52 / 52.93	98.05 / 97.85	-0.59 [-2.34, 1.17]
Qwen3-14B	63.09 / 60.55	97.46 / 97.85	-2.54 [-5.47, 0.20]

Table 23: Complete supplementary readouts; Confirmation model.

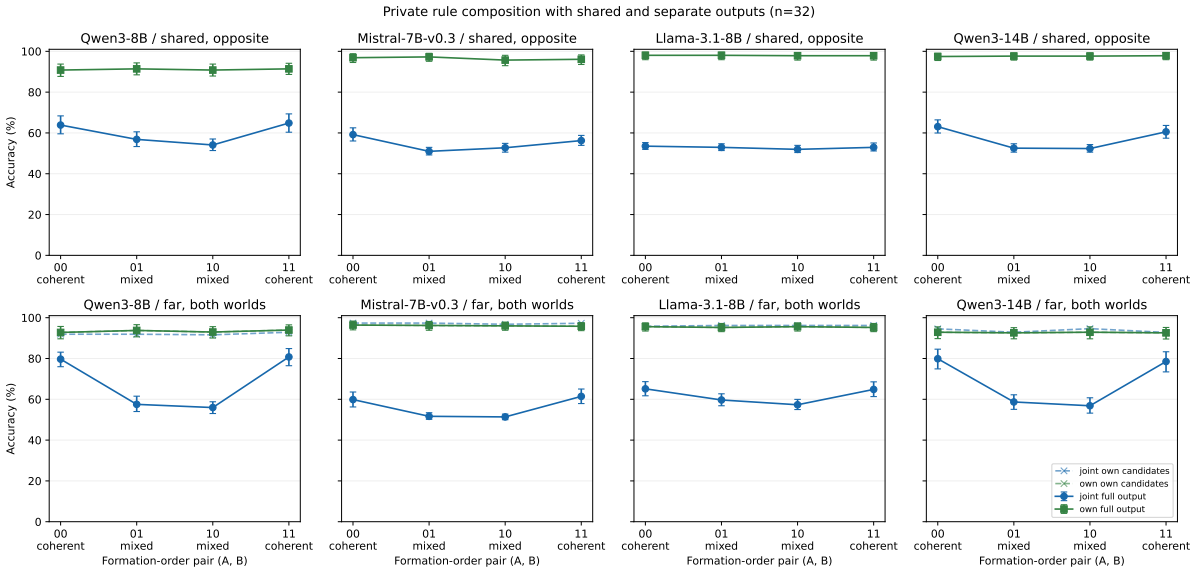


Figure 12: All cache pairs and private controls

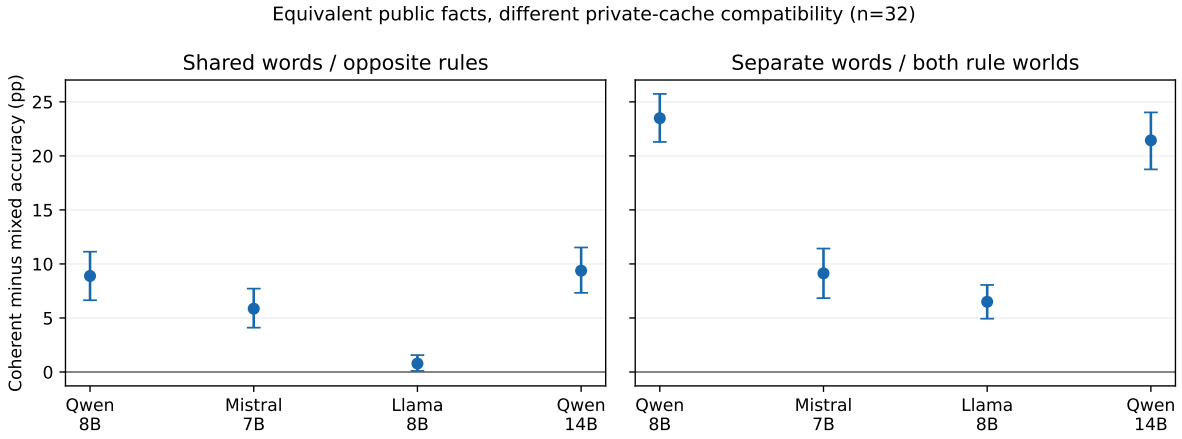


Figure 13: Shared-word rule use and separate-word use

Model / phase	Primary coherence, pp [95% CI]	Criterion-oriented margin coherence T, nats [95% CI]
Qwen3-8B / discovery	2.34 [0.39, 4.69]	0.03660 [0.01877, 0.05657]
Mistral-7B-v0.3 / discovery	-0.78 [-1.95, 0.00]	0.00853 [0.00412, 0.01442]
Llama-3.1-8B / discovery	1.56 [0.00, 3.52]	0.00767 [0.00501, 0.01085]
Qwen3-14B / discovery	0.00 [-1.95, 2.34]	0.04048 [0.02804, 0.05418]
Qwen3-8B / confirmation	1.56 [-0.59, 3.52]	0.04256 [0.03058, 0.05624]
Mistral-7B-v0.3 / confirmation	0.98 [-1.37, 3.32]	0.00502 [0.00303, 0.00714]
Llama-3.1-8B / confirmation	0.78 [0.00, 1.76]	0.00837 [0.00624, 0.01068]
Qwen3-14B / confirmation	2.93 [0.59, 5.47]	0.05037 [0.03974, 0.06198]

Table 24: Complete supplementary readouts; Model / phase.

Confirmation model	Joint00 / 11, % [95% CI]	Own-only00 / 11, % [95% CI]	Single Food / Service on conflicting queries, % [95% CI]
Qwen3-8B	50.78 [48.05, 53.91] / 52.73 [49.61, 55.86]	56.25 [51.95, 60.55] / 58.20 [53.91, 62.89]	46.88 [40.62, 52.34] / 69.53 [62.50, 77.34]
Mistral-7B-v0.3	50.39 [48.05, 52.73] / 50.00 [48.05, 51.95]	54.69 [51.95, 57.81] / 55.47 [51.95, 59.38]	57.81 [53.12, 64.06] / 53.12 [49.22, 57.81]
Llama-3.1-8B	51.17 [50.00, 52.73] / 51.95 [50.39, 53.52]	70.70 [64.84, 76.95] / 70.31 [64.45, 76.56]	67.19 [60.16, 74.22] / 71.09 [63.28, 78.91]
Qwen3-14B	52.34 [48.83, 55.86] / 51.95 [49.61, 54.30]	75.00 [71.09, 78.91] / 73.05 [69.14, 76.95]	68.75 [61.72, 75.80] / 78.12 [70.31, 85.94]

Table 25: Complete supplementary readouts; Confirmation model.

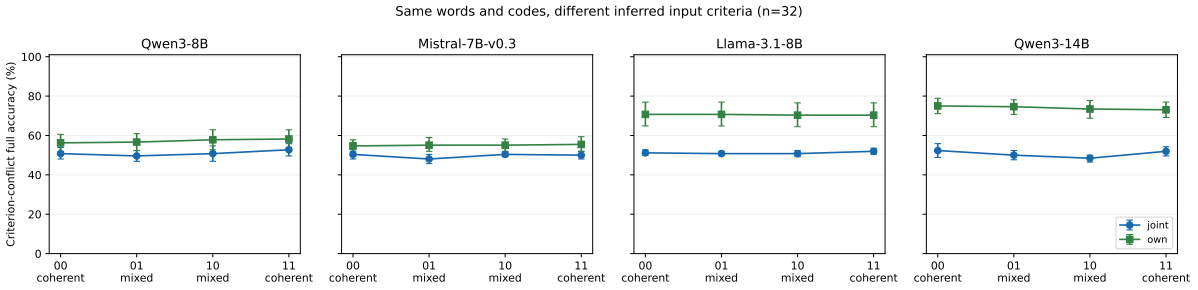


Figure 14: Criterion-conflict cache compatibility

Model / phase	CueB det, nats² [95% CI]	Source-specific det, nats² [95% CI]
Llama-3.1-8B / discovery	-5.914e-06 [-1.213e-05, 5.864e-07]	6.157e-06 [2.463e-06, 1.080e-05]
Qwen3-14B / discovery	-9.656e-06 [-1.251e-04, 8.155e-05]	5.551e-06 [-1.524e-05, 2.613e-05]
Llama-3.1-8B / confirmation	2.492e-06 [-5.264e-06, 1.097e-05]	-1.259e-06 [-3.316e-06, 4.967e-07]
Qwen3-14B / confirmation	3.740e-05 [-1.643e-05, 1.014e-04]	-1.148e-05 [-4.553e-05, 1.924e-05]

Table 26: Complete supplementary readouts; Model / phase.

Confirmation model	Single Food early / late conflict accuracy, % [95% CI]	Single Service early / late conflict accuracy, % [95% CI]	Native B early / late origin conflict accuracy, % [95% CI]
Llama-3.1-8B	68.75 [60.94, 76.56] / 69.53 [62.50, 77.34]	71.09 [64.84, 77.34] / 74.22 [67.97, 80.47]	60.94 [51.56, 69.53] / 63.28 [55.47, 71.09]
Qwen3-14B	78.12 [70.31, 85.16] / 78.91 [71.88, 85.94]	76.56 [68.75, 84.38] / 79.69 [71.88, 87.50]	67.19 [60.94, 74.22] / 64.84 [56.25, 73.44]

Table 27: Complete supplementary readouts; Confirmation model.

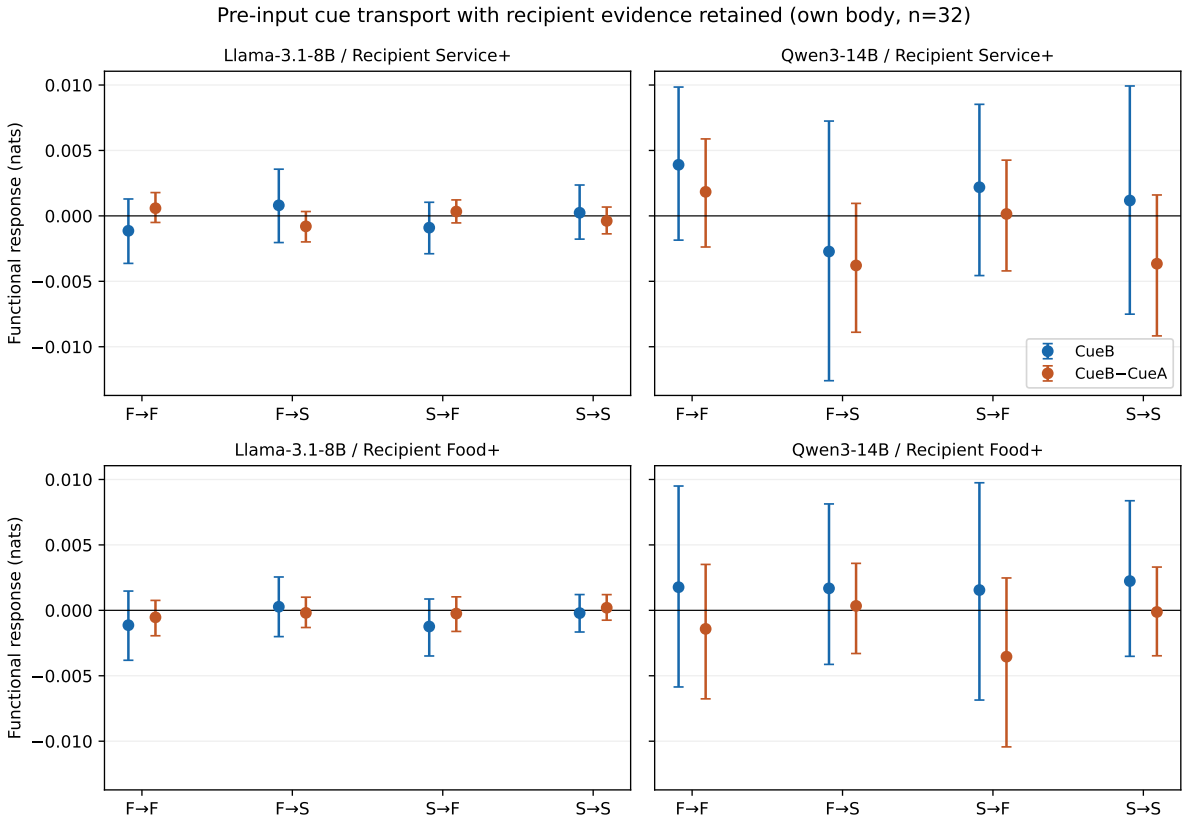


Figure 15: Functional cue information

Confirmation model	Native CueB det, nats² [95% CI]	Native Source-specific det, nats² [95% CI]
Llama-3.1-8B	−1.751e−5 [−4.604e−5,3.566e−6]	2.940e−6 [−2.280e−5,3.356e−5]
Qwen3-14B	−1.821e−4 [−6.037e−4,2.459e−4]	−3.650e−4 [−9.933e−4,2.110e−4]

Table 28: Complete supplementary readouts; Confirmation model.

Qwen3, fresh confirmation n32	Full-source answer removal, nats	A-only restricted restoration, nats	Full-source restricted restoration, nats
8B	1.762 [1.264, 2.294]	2.143 [1.426, 2.893]	−.650 [−1.269, −.080]
32B	1.972 [1.467, 2.500]	.775 [.321, 1.284]	−.544 [−.833, −.276]

Table 29: Complete supplementary readouts; Qwen3, fresh confirmation n32.